

# The effects of leaving the EU on the geography of UK trade

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## Abstract

This paper investigates the effect of the 2016 Brexit referendum and the Trade and Cooperation Agreement (TCA) signed by the UK and the EU in December 2020 on UK-EU trade in goods up to December 2021. The 2016 referendum introduced uncertainty, but new trade barriers between the UK and the EU were not put in place until the new agreement, the TCA, entered into force in January 2021. Using a set of different econometric techniques, and looking up to the end of the first year of the implementation of the TCA, we find no evidence that the 2016 referendum had any impact on aggregate UK-EU trade relative to comparator trade flows. On the other hand, our results show that the TCA reduced UK trade with the EU, but asymmetrically for exports and imports. After a sharp drop of around 41% in January 2021, UK exports to the EU quickly recovered in the following months. This pattern of recovery was not the case for UK imports from the EU, which were negatively impacted throughout of 2021, with a cumulative loss over the first year of implementation between -24% and -28%. Asymmetric effects between exports and imports do not appear to be driven by difference in product composition of the two flows. Instead, we find evidence that a plausible explanation for these differential effects is the relative importance of the EU market for UK firms, in comparison to the importance of the UK market for EU firms.

## 1 Introduction

In January 2021, after a last-minute deal on Christmas Eve 2020, the UK and the EU entered into new trade arrangements governed by the Trade and Cooperation Agreement (TCA). While the TCA maintains tariff-free trade between the two signatory parties, leaving the EU's Single Market resulted in the introduction of a range of non-tariff barriers, from customs procedures to regulatory barriers. These result in an increase in the costs of bilateral UK-EU trade. The fact that the agreement was signed at the very last minute also meant that there was uncertainty for firms on both sides of the English Channel prior to January 2021 as to the conditions under which trade with the EU would occur. The uncertainty started immediately after the referendum in June 2016 (if not before), and then fluctuated depending on the state of play of the negotiations. January 2021 saw the introduction of the new trading arrangements and asymmetric customs formalities between the UK and the EU.

In this paper we assess the impact of the 2016 Brexit referendum itself and the introduction of the TCA on the geography of UK trade. Specifically, we examine the impact on trade between the UK and the 27 EU member states with monthly data up to December 2021. We do so comparing UK-EU trade to other trade flows as detailed in the methodological section. Estimation of the impacts of the TCA effect is made challenging by two concurring issues: the Covid-19 pandemic which disrupted economies, and the sharp rise in shipping

cost – which were also in large part a consequence of Covid-19.<sup>1</sup> Similarly, the analysis of the effect of the referendum might be affected by other confounders. In order to isolate the impact of the referendum and of the TCA from other forces we apply a series of econometric techniques, which are difference-in-differences (DD), triple difference (DDD) and synthetic control. In our principal regressions we use total trade flows. It is possible that Covid may have had different impacts on different countries' trade; and/or that there may have been a differential impact on trade in Covid-related products which thus affects our aggregate regressions; and/or that Covid may have impacted differentially across sectors also affecting our aggregate regression. As discussed in Section 2 on the empirical strategy we control and test for each of these possibilities.

It is important to note that in this paper we do not examine the impact on total UK trade to all destinations. Our focus is on UK trade with the EU. This is because, *prima facie*, if Brexit were to impact on trade, in the first instance we would expect this to impact on relative trade with the EU. Nevertheless the aggregate trade effects are clearly of interest.

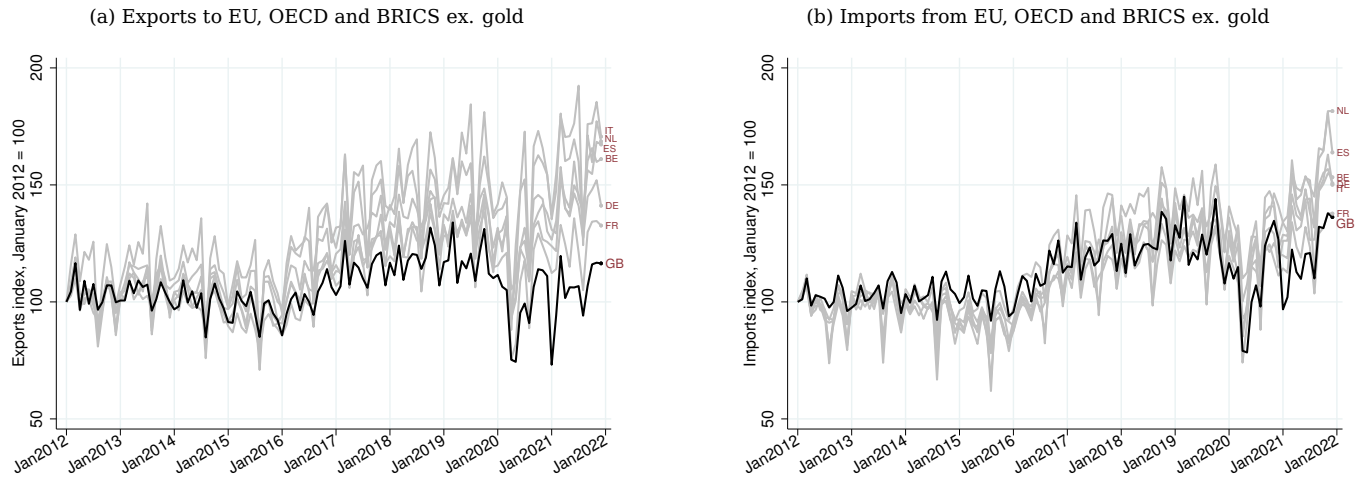
Before we turn to our formal analysis, Figure 1 shows the evolution of trade excluding gold of the UK and the other major EU countries with the EU, OECD and BRICS.<sup>2</sup> We can immediately see that, while prior to 2021 UK trade followed a very similar trajectory to those of the other major EU members, in 2021 UK trade has underperformed compared to these other countries. The same is true for UK GDP. Figure 2 plots the quarterly GDP volume index (2015=100) for all OECD countries. Among the OECD countries, the UK is the one that suffered the most when the pandemic hit in 2020q1. Moreover, by 2021q3 UK's GDP had not recovered to pre-pandemic levels and only four OECD countries performed worse than the UK at this stage. With respect to goods trade, we will see that a large part of the UK's under-performance is attributable to trade with the EU, which dropped significantly with the introduction of the TCA and accounts for a large share of total UK trade in goods, and this is particularly true for imports.

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<sup>1</sup>There is a third issue which concerns changes in the way trade flows are recorded from January 2021. See Gasiorek and Tambari (2021) for an extensive discussion of data concerns.

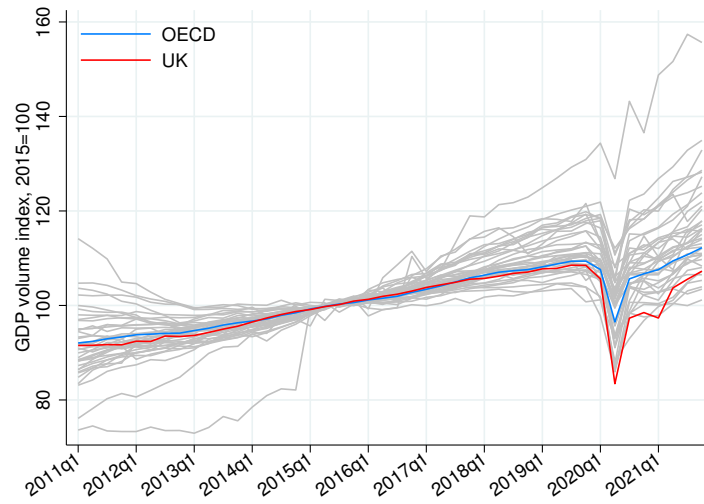
<sup>2</sup>We exclude gold, as trade in gold can fluctuate substantially on a month-to-month basis and thus impact on the aggregate figures.

Figure 1: Trade of the UK and major EU countries, Jan2012-Dec2021



Source: authors' elaboration of HRMC and Eurostat data. UK trade data are taken from HMRC trade statistics, while trade data of EU member states come from the Eurostat Comext database.

Figure 2: GDP volume index for OECD countries, 2015=100



Source: authors' elaboration of OECD data.

Coming to a description of the methods used in this paper, in the difference-in-differences (DD) exercise we compare UK-EU trade with UK trade with a suitable control group, which is composed of non-EU OECD

countries plus BRICS.<sup>3</sup> In this setting, our assumption is that in absence of the referendum or the TCA, UK trade with the EU would have followed the same trajectory as UK trade with the control group. In the estimation of the TCA effect, the DD allows us to control for the evolution of Covid-19 in the UK. For robustness we also control for the evolution of the pandemic in other countries including as variables the number of Covid-19 cases and deaths in each trade partner. For the triple difference estimation (DDD) we also include information about EU trade with itself and with the countries in the control group. In this way we are able to control for factors which are specific to each country and period, such as the evolution of Covid-19, the implementation of lock-downs and any economic shock that affected a country in a particular period. Finally, we also recognise the possibility that not all countries in the control group may be a good comparison unit for UK-EU trade. For instance, Germany-France trade might be a better counterfactual for Germany-UK than say US-Mexico. We therefore use the Synthetic Control (SC) method, which selects as control units the subset of countries which represent the best match for UK-EU (details below). The estimation with aggregate data does not deal with possible heterogeneous effects across products. We therefore estimate the model at a more disaggregated level and find evidence of heterogeneous effects across products, especially for UK exports to the EU. However, when we compute the treatment effects for total UK-EU trade aggregating the product-level estimates we find very similar results to our estimates based on aggregate trade flows.

The results appear very stable across the different methodologies. We do not find any consistent evidence that the 2016 referendum had a sizeable impact on aggregate UK-EU trade, neither for exports nor for imports. On the other hand, we find strong negative effects of the TCA on UK-EU trade, in particular for UK imports from the EU. The results show that UK exports to the EU were strongly affected in January (between -41% and -43%) but quickly recovered thereafter, suggesting that this sharp drop was mainly driven by the lack of time firms had to prepare for the eventual agreement. Over the period January-December 2021, the cumulative effect on exports was between -4% and -11.8%, hardly statistically significant. However, while this is true in aggregate, we also find that some sectors have been strongly affected over the entire period, in particular Textile & Clothing and Footwear. On the other hand, we find that UK imports from the EU were consistently affected in all the months of 2021, with a cumulative effect of -24% to -28% over January-December 2021. Moreover, we find some evidence of anticipatory effects on UK imports from the EU, which started to decline a few months before the implementation of the TCA. The differences across sectors for imports are modest compared to those found with regard to exports. The results on imports are perhaps, *prima facie*, surprising given that the UK waived most of the border checks on imports from the EU, while EU members immediately began enforcing checks on UK goods.<sup>4</sup>

One possible explanation for the asymmetric imports and exports effect can be product composition. If UK exports are concentrated in products which saw less increase in trade barriers while the opposite is true for EU exports, then we could find the pattern of results that we observed. We test this hypothesis by looking at trade disaggregated by product but find no support for it: even at the HS 2-digit products (96 products) the exports and imports effect remain highly asymmetric.

Given that the EU is a larger market for UK exports than the UK is for EU exports, we ask whether relative market sizes might be a driver of the different results for exports and imports. We test this using a two-step approach involving the estimation TCA effects for each HS section and EU27 member. Our empirical test fails to reject the market size hypothesis, providing evidence in support of the idea that the differential results

<sup>3</sup>From this control group we also exclude Canada and Japan as they entered into FTA with the EU over the considered period.

<sup>4</sup>While the EU imposed full customs controls on UK exports to the EU, in contrast the UK decided to delay the introduction of full customs controls until later in the year. In the event, those dates also were moved and the UK introduced full customs controls from January 2022, with a further staging of controls and inspections for food and animal products till mid-2022, and later for some products.

between exports and imports can be explained by relative market sizes.

## 1.1 Policy background

As an EU member state the UK was part of the EU's Customs Union and part of the EU Single Market. The Single Market applies to countries in the European Economic Area (EEA), which includes the EU as well as Iceland, Liechtenstein and Norway. Leaving the EU entailed a departure from each of these for the UK, but differentially for Great Britain and for Northern Ireland. Hence, Great Britain left both the Customs Union and the Single Market. In contrast Northern Ireland remains in the EU's Single Market for goods, but not for services, and while not part of the Customs Union remains part of the EU's common customs territory. The departure from the CU and the SM have impacted on the degree of market access between the UK and the EU (and the UK and third countries) in different ways. The changes in market access and the increase in firms' costs thus involves a combination of tariff barriers, administrative barriers, and regulatory barriers.

Departure from the Customs Union means that the UK is free to set its own most favoured nation (MFN) tariffs on imported goods and is able to sign free trade agreements with third countries. The UK's MFN tariff was established in May 2020, and is commonly referred to as the UK Global Tariff (UKGT). To date the UK has also managed to sign new free trade agreements with the EU (2021), Japan (2020), Australia (2022) and New Zealand (2022). The UK has also signed a series of 'continuity agreements' with those countries which have a free trade agreement with the EU and consequently as did the UK when it was a member of the EU. The aim of the continuity agreements is to replicate as closely as possible the previous arrangements.

The agreement with the EU in principle abolishes all tariffs on trade between the UK and the EU. 'In principle' is highlighted here because importantly tariff free trade only applies to goods which originate in the UK and the EU respectively, and which can be shown to originate. In order for a good to be deemed as originating there has to be sufficient economic activity undertaken in the originating country (either the UK or the EU), and there are complex 'rules of origin' for each product detailing the minimum origin requirements. In exporting to the EU, therefore, if a given UK firm wants tariff free access it is required to be able to prove that the good is eligible for duty free access. In order to reduce the barriers to market access the TCA allowed for either self-certification of origin by the exporter, or self-certification by the importer. In addition and for the first 12 months the TCA did not require firms to provide the supporting documentation at the time of import, though that documentation for each flow could be subsequently demanded. All of this serves to increase market access costs for firms because of the greater need to track their supply chain in order to know whether or not they are eligible for preferential access, as well as the administrative and bureaucratic costs of providing the necessary certificates of origin, and of ensuring they have the required documentation for verification of this. The current evidence on the extent to which UK firms' are exporting using the preferential zero tariff rates, suggests that over the first year of the TCA on average the preference utilisation rate was only 76%, and in some sectors such as leather goods (HS 41-13) as low as 33%.<sup>5</sup>

Leaving aside the rules of origin issue, all goods traded between the UK and the EU now face customs formalities and potentially customs checks where previously this had not been the case, such as the completion of import or export declarations. For some goods the customs documentation and procedure may be relatively light, but for others and notably those subject to sanitary and phytosanitary requirements there are a range of procedures. These include, import pre-notifications, health certification (either an Export Health Certificate

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<sup>5</sup>Shares based on EU27 imports from the UK in 2021 under Normal statistical regime. For an overview of preference utilisation across sectors up to July 2021 see Ayele, Larbalestier and Tamberi (2021).

or Phytosanitary Certificate), documentary, identity and physical checks at the border, and entry into the partner market via a Border Control Post (BCP).

The preceding highlights that leaving the EU Customs Union led to the possibility of tariffs on some trade between the UK and the EU. In addition, the departure of Great Britain from the EU's Single Market introduces further regulatory barriers to bilateral trade.<sup>6</sup>

A key principle of the Single Market is the mutual recognition of goods and services. Mutual recognition in the Single Market means that where rules are not harmonised at the EEA level, EEA Member States must recognise each other's regulations (except where there are specific derogations). Once a product is lawfully placed on the market in one Member State, it can be marketed in another Member State without barriers. A key issue here, however, is not simply with regard to whether or not the UK and the EU have the same standards and whether or not they recognise each other standards, but concerns the arrangements for proving conformity to those standards. In a trade agreement between any pair of countries this could be addressed via an agreement on the mutual recognition of conformity assessment. This would acknowledge the differences between regulatory regimes but permits one party to test and certify that a product complies with the other party's regulations. In the UK-EU TCA there is no chapter on the mutual recognition of conformity assessment. There are some sectoral exceptions, and for least sensitive products 'self-certification' is possible, but for the remaining products goods made in the UK for sale in the EU must not only conform to EU standards but they must provide EU-overseen paperwork to prove this conformity. Even where 'self-certification' is allowed, the firm will have to appoint an EU-based representative who will take legal liability if necessary. For goods which require third party certification the testing has to be carried out in the EU by an EU accredited body.

Consider Figure 3 below, which is based on taking all the EU Single Market regulations and directives which impose some form of obligation on firms - be this with regard to product standards, process standards, labelling or marking requirements, or with regard to conformity assessment.<sup>7</sup> For each regulation the HS 6-digit products to which the regulation applies have been identified. At the 6-digit level there are close to 2,000 products for which five or more EU regulations/directives apply. The chart below, then gives the number of regulations that apply at the more aggregate HS 2-digit level. This provides some indication of the degree of regulatory intensity. This ranges from 46 regulations in HS03 (fishing), and 46 regulations in HS84 (machinery and mechanical appliances), to only 2 regulation in HS92 (musical instruments).

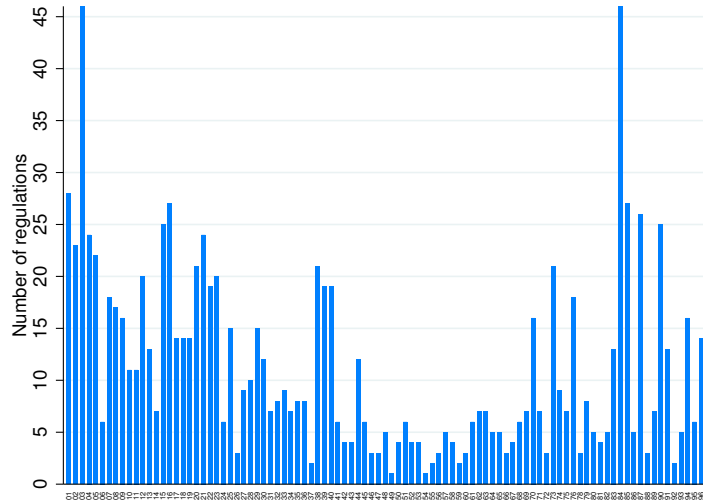
Of course, the presence of a mandatory standard (regulation or directive) does not necessarily entail a barrier to market access. Indeed, being able to produce to a given standard may well facilitate market access. Nevertheless, it is the case that firms exporting to the EU have to conform to these regulations and directives, and have to be able to prove that they conform. The higher the number of regulations that need to be satisfied the more likely it is that the regulatory burden will be higher.

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<sup>6</sup>The United Kingdom comprises Great Britain (England, Scotland and Wales), plus Northern Ireland. Northern Ireland has remained in the EU's Single Market for goods.

<sup>7</sup>This work derives from an ongoing ESRC funded, Governance After Brexit Project, entitled 'Brexit Uncertainty and the Northern Ireland Protocol: The consequences for Northern Ireland firms and their trade within the UK's Internal Market', led by Professors Gasiorek and Reilly.

Figure 3: Single Market regulatory intensity: The number of regulations at the HS 2-digit level



Source: authors' calculation.

## 1.2 Literature review

The departure of the UK from the EU, as a process of trade policy ‘disintegration’ is a somewhat unprecedented policy change, which thus limits the existing relevant empirical literature. Regarding the possible impact on the UK, there is small literature on the impact of the Brexit referendum on UK-EU trade, mostly focusing on the effect of uncertainty. Crowley, Exton and Han (2018) show that firms’ entry into exporting to the EU was reduced in the first six months after the referendum as a consequence of an MFN tariff threat. Graziano, Handley and Limão (2021) use odds from the betting market to track the evolution of the Brexit referendum uncertainty over time, and show that the MFN threat started to affect UK-EU trade even before the referendum of June 2016. Looking at aggregate bilateral trade between the UK and 14 EU member states, Douch and Edwards (2021) apply the synthetic control method and find that UK-EU trade started to decline after the 2015 general election won by the Conservative party. Fernandes and Winters (2021) look at the response of Portuguese exporters to the UK following the 2016 referendum. Comparing firm-level exports to the UK and other destinations, the authors find a decrease in exports prices and volumes to the UK after the referendum, with heterogeneous effects along exporters and products. Work by Nardotto and Sequeira (2021) using a large data set from a major UK retailer finds that following Brexit there was a switch of demand by UK consumers, with a 6% increase towards UK produced goods, and a reduction of 13% away from EU goods.

There are also a series of ex-ante simulation studies that estimate the potential impact of UK exit from the EU under different scenarios. Here we report the results of some of the main ex-ante simulation papers without aiming to a comprehensive literature review on simulated Brexit impacts.<sup>8</sup> In terms of UK-EU trade effects, the static general equilibrium ex-ante modelling of Dhingra et al. (2017) gave a short-run (1 year) reduction by -14% for UK exports to the EU and -13% for UK imports from the EU under soft Brexit scenario,

<sup>8</sup>Many ex-ante simulation focus on GDP and welfare effects, and not all studies report effects on bilateral trade.

and -36% for exports and -34% for imports under WTO scenario (see Table 5 in Dhingra et al. (2017)). Steinberg (2019) uses a dynamic general equilibrium model and simulates both the effects of referendum-induced uncertainty and actual exit from the EU on UK macroeconomic variables. The author finds little effects of uncertainty on the UK economy, with negligible effects on trade, but substantial negative effects arising with the actual exit from the EU. Under the soft (hard) Brexit scenario exports to the EU fall by -7.7% (-31.8%) and imports from the EU by -10.4% (-44.1%).<sup>9</sup> Felbermayr, Gröschl and Steininger (2021) use a gravity model on WIOD data to estimate the impact of EU membership and then use those estimates to simulate different Brexit scenarios. In the FTA (WTO) scenario, UK exports to the EU in Manufacturing change by +7.16% (-32.19%) while imports decrease by -4.71% (-30.63%). It is important to recognise that most ex-ante results are based on a mix of increase in tariff and non-tariff barriers. The TCA maintained tariff-free trade – conditional on meeting origin requirement – and mainly increased non-tariff barriers. Hence, the quality of ex-ante studies will depend on the accuracy with which they can measure non-tariff barriers, which are generally difficult to quantify.

Examples of trade disintegration are hard to find. Daigle, DeCarlo and LaRocca (2019) take the period 1949-2019 and identify more than 163 examples of the ending of trade arrangements between countries, most of these relate to countries (especially former Eastern European countries) leaving an agreement either to improve on an existing bilateral agreement, or to join a more liberalized and/or larger trading block. The majority of cases of countries leaving trade agreements without switching to an alternative agreement have occurred in Africa where the circumstances are not comparable to the UK. Other studies of the ending of free trade agreements include Gnutzmann, Gnutzmann-Mkrtchyan and Korn (2021) who explored the impact of Estonia leaving an FTA with Ukraine in order to join the EU using a difference-in-difference framework; Fidrmuc and Fidrmuc (2003), who used a gravity model to assess the consequences of the disintegration of the former Soviet Union, Yugoslavia, and Czechoslovakia on trade flows over the period 1990-98; and De Ménil and Maurel (1994) who also applied the gravity framework to look at the dissolution of the Austrian-Hungarian empire in the period 1924-26. None of these are particularly informative as examples for the consequences of the UK's departure from the EU.

Another instance of trade policy reversion which has been studied more extensively recently was the introduction of tariffs by the US, and particularly the Trump administration. These were either focussed on specific sectors (initially solar panels and washing machines and subsequently steel and aluminium) or on trade with China. In particular from 2018, the US administration imposed significantly higher tariffs on specific products and countries on imported goods worth more than \$300 billion. In response, several countries including China, the EU, Canada, Mexico, and Turkey also imposed retaliatory tariffs on their imports from the US. Amiti, Redding and Weinstein (2019) find that the Trump tariffs increase led to substantial increase in the price of final and intermediate goods and a reduction in the number of imported varieties. The authors, in line with research of Cavallo et al. (2021) and Fajgelbaum et al. (2020), find that the increase tariffs has been entirely passed through into domestic prices.

While clearly an example of a trade policy reversal the US example is also a poor proxy for the departure of the UK from the EU. This is for at least two reasons. First, because the shares of trade involved are considerably smaller. As documented by Fajgelbaum et al. (2020) the US tariffs were imposed on 12,043 national 10-digit products covering \$303 billion, amounting to 12.7% of annual U.S. imports. The retaliatory tariffs on U.S. exports by its trade partners covered 8,073 export products, worth \$127 billion, which amounted to 8.2% of the annual U.S. exports. This is considerably less than the amount of trade between the UK and the

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<sup>9</sup>Steinberg 2019 reports results for trade with the EU as a percentage of GDP in Figure 4. We derived the impacts on trade flows dividing the changes of Figure 4 by those for GDP reported in Figure 3 of Steinberg (2019).



EU. In 2020 the share of UK exports going to the EU was 47.5% and the share of imports coming from the EU was 53.2%.

Secondly, the actions of the US administration were focussed on tariffs and on trade in goods. The departure of the UK from the EU came with a tariff free and quota free trade agreement (providing rules of origin can be met), and the rise in barriers was much more to do with customs formalities and behind the border regulatory changes arising from the UK's departure not just from the EU Customs Union, but also from the Single Market. Importantly too, leaving the Single Market increased the barriers to trade in both goods and services. One aspect of that increase in market access barriers relates to the introduction of rules of origin on trade between the UK and the EU. Empirical work on the impacts of rules of origin and cumulation schemes is also relatively scarce. This arises from the complexity of the underlying rules and to the methodological difficulties of isolating the impact. Augier, Gasiorek and Lai-Tong (2004) and Augier, Gasiorek and Lai Tong (2005) provide an empirical assessment using the gravity framework of rules of origin in the context of the pan-European system of diagonal cumulation introduced in 1997. They find that reducing the restrictiveness of rules of origin (via the introduction of diagonal cumulation) decreased trade both between the EU's partner countries and those partners' trade with third countries by between 7.4% to 22.1%. In the case of the UK and the EU, the situation is one of introducing rules of origin on bilateral trade, and a reduction in the extent to which both parties can apply diagonal cumulation. Carrère and De Melo (2004) use data on the preference utilization rates of Mexican exports to the USA in 2001 to estimate the costs of different rules of origin for final and intermediate goods. They find that the costs of different rules of origin vary between 2.3% and 4.6% depending on the type of rule of origin and whether or not it is applied to final or intermediate goods. Focusing on the North American Free Trade Agreement (NAFTA), Conconi et al. (2018) match input-output linkages to NAFTA's Rules of Origins and show that the agreement substantially reduced imports of intermediate products from third countries relative to NAFTA signatories. In more recent work Chung and Perroni (2021) assess the impact of rules of origin on competition and prices, based on the 1989 Canada-United States free trade agreement (CUSFTA). They find that stricter content requirements are associated with higher markups and more (inefficient) firm entry that could then hinder pro-competitive effects of trade liberalisation.

## 2 Econometric method

We use three different methods to measure the impact of the 2016 Brexit referendum and the TCA on UK EU trade. These are difference-in-differences (DD), triple differences (DDD) and synthetic control (SC). The rest of this section describes the methodologies in details. As in any empirical investigation international trade data, our analysis will not be able to retrieve levels effects of the TCA. Instead, our focus is on the geography of trade estimating relative trade effects. Because the TCA imposes trade barriers between the UK and the EU, we believe that the geography of trade is an important dimension to study if we want to find causal effects of the TCA.

### 2.1 Difference in differences

For the DD method we estimate the following equation with Poisson PML:

$$Y_{jt} = \exp [\alpha_j + \alpha_t + \beta (EU_j \times Post_t)] + \epsilon_{jt} \quad (1)$$

where  $Y_{jt}$  is either UK exports to or imports from partner country  $j$  in period  $t$ ,  $\alpha_j$  and  $\alpha_t$  are partner and time dummies respectively,  $EU_j$  is a dummy that takes value of one if the partner  $j$  is an EU member and  $Post_t$  is a dummy that equals one if period  $t$  is in the post-treatment period. For the referendum, the pre-treatment period is Jan2012-Jun2016, while for the TCA the pre-treatment is Jan2017-Dec2020.

The parameter of interest is  $\beta$ , which measures the effect of the referendum or TCA on UK-EU trade relative to UK trade with the control group, and it is equivalent to the ratio of ratios  $\beta = \frac{\bar{Y}_{EU,Post}}{\bar{Y}_{EU,Pre}} / \frac{\bar{Y}_{control,Post}}{\bar{Y}_{control,Pre}}$ , where  $\bar{Y}$  indicates the average.

We use the PPML estimator rather than the OLS on a log-linear version of equation (1) because with PPML we can estimate the proportional effect of the TCA on total UK-EU trade even with a dataset disaggregated by partner countries. In presence of treatment heterogeneity across EU countries, the PPML estimates a weighted treatment effect with weights given by the size of the trade flow with each country. This because a diff-in-diff estimated by PPML gives the proportional effect on the *arithmetic* average of the dependent variable  $Y$ . On the other hand, a diff-in-diff estimated with a log-linear OLS would give the proportional effect on the *geometric* average of  $Y$ , hence we would estimate an average effect across countries, not the effect on the average. More details can be found in the Appendix (see Section A.1. See also Breinlich, Novy and Santos Silva (2022) on aggregation with the PPML estimator).

The second DD specification that we use to estimate the TCA effect in each month of the 2021 is:

$$Y_{jt} = \exp \left[ \alpha_{j,month} + \alpha_t + \sum_{month} \beta_{month} (EU_j \times D_{t,month}^{2021}) \right] + \epsilon_{jt} \quad (2)$$

where  $\alpha_{j,month}$  now is a dummy for each partner-month (i.e., one for Germany-January, one for Germany-February, ...) and the variables  $D_{t,month}^{2021}$  are dummies for each month of 2021. With this setting the coefficients  $\beta_{month}$  measure the TCA effect for each month of 2021 relative to the average of the same month over the period 2017-20 accounting for seasonality (e.g., the  $\beta_{Jan}$  coefficient measures the TCA effect in January 2021 relative to the average of Januaries 2017-20). For the referendum we estimate a similar equation but estimating effects for each half years.<sup>10</sup>

## 2.2 Triple difference

For the triple difference estimation we expand our dataset to include information on trade of each EU member state with other EU members and also EU members trade with countries in the control group. The triple difference equation for the TCA effect is:

$$Y_{ijt} = \exp [\alpha_{it} + \alpha_{jt} + \alpha_{ij} + \beta (EU_j \times UK_i \times Post_t)] + \epsilon_{ijt} \quad (3)$$

where  $Y_{ijt}$  is trade (exports or imports) of reporter  $i$  (UK and EU27) to partner  $j$  (EU27, OECD+BRICS) in period  $t$ . The cross-section unit is now the country pair  $ij$  and  $\alpha_{it}$ ,  $\alpha_{jt}$  and  $\alpha_{ij}$  are reporter-time, partner-time and pair dummies, respectively. Finally,  $EU_j$  is a dummy that equals one if partner  $j$  is an EU member state and  $UK_i$  is a dummy that equals one if reporter  $i$  is the UK. Strictly speaking, this specification is not the classic triple difference and it is more similar to a gravity model – but it is also not a full gravity model.<sup>11</sup> The estimation of the TCA effect  $\beta$  in this model is the average of two triple differences: one within reporter,

<sup>10</sup>Given that the post-referendum period is relatively long covering Jul2016-Dec2019, half-year estimates are easier to present. In unreported results we also estimated month-specific effects for the post-referendum period.

<sup>11</sup>We do not use a complete trade matrix dataset because monthly bilateral trade data are not readily available for all countries.

and one within partner. For exports, the within reporter DDD (given by the inclusion of  $\alpha_{it}$  dummies) is the comparison of UK to EU vs UK to control (first difference) before and after 2021 (second difference) with the same quantity for EU to EU vs EU to control (third difference). The second DDD is within partner: EU from UK vs EU from EU (first difference) before and after 2021 (second difference) vs the same quantity for control from UK and from EU (third difference). Moreover, because of the PPML estimator these are not really ‘difference in differences’ but ‘ratio of ratios’. The difference with the gravity model is that gravity includes trade between units in the control group and adds further comparisons such as UK to France vs USA to Mexico. To estimate the DDD model and compute month-specific effects of the TCA we estimate:

$$Y_{ijt} = \exp \left[ \alpha_{it} + \alpha_{jt} + \alpha_{ij,month} + \sum_{month} \beta_{month} (EU_j \times UK_i \times D_{t,month}^{2021}) \right] + \epsilon_{ijt} \quad (4)$$

where  $\alpha_{ij,month}$  are pair-month dummies (that is, UK-Germany-January, UK-Germany-February, ..., France-Germany-January etc). Similarly to the DD equation, we estimate half-year post-referendum effects using the triple-difference specification. We acknowledge that some anticipation effects might be occurring in the months preceding the implementation of the TCA. To see how important they are, we also estimate an event study specification (see Section 4.4).

As widely reported in the news, the cost of shipping goods around the world increased dramatically in 2021, and these changes might affect our results. To control for the effect of shipping cost we make the assumption that these costs are proportional to the distance between two trading partners and use an interaction of the log of distance with a time-varying measure of shipping cost  $Freight_t$  (details on how we measure this variable below). Our assumption here is that the increase in freight costs is proportional to distance between partners, with time variation common across all trading countries. Note that we would not want to include pair-specific trade costs that vary over time because, for UK-EU, this will be endogenous to the introduction of the TCA. As a second specification, since our cost of shipping variable is common across all partners and varies only in time, we then replace the shipping cost variable with time dummies and interact them with the log of distance.

## 2.3 Synthetic control

The last method that we apply is the synthetic control developed by Abadie and Gardeazabal (2003) and Abadie, Diamond and Hainmueller (2010). The exposition is done from the perspective of the TCA estimation, but we also run the SC for the referendum period. For the synthetic control our dataset is the same as the one used for the DDD estimation. In total there are  $27 \times 2 = 54$  treated units: UK exports to each of the 27 EU members and UK imports from each EU member. We run the SC for each treated unit separately, and for each treated unit the donor pool is composed of UK’s trade with the control group and the EU member’s trade with the control group and other EU members. For instance, if the treated flow is UK exports to Germany the donor pool includes UK exports to control (UK to US, UK to Australia ...), Germany imports from control group (e.g., Germany from US) and Germany imports from other EU members (e.g., Germany from France). In our main specification we match over all pre-TCA values of the outcome variable (July 2016-December 2020), which as discussed in Ferman, Pinto and Possebom (2020) reduces the possibility of specification search with synthetic control. We also run as robustness a specification in which we match over the pre-treatment averages of each month (i.e., average of Januarys, of Februarys...over the period July 2016-December 2020).

Inference is based on the permutation method. For each treated unit we run the SC for all other 27 EU members so to have 27 placebos. So for UK to Germany the placebos are France to Germany, Italy to Germany etc. First, we compute the normalised root mean squared prediction error (RMSPE) for each

month of the post-treatment period, where the normalisation is division by the pre-treatment fit as suggested in Abadie, Diamond and Hainmueller (2010) and Abadie (2021):

$$r_{it} = \frac{\sqrt{(Y_{it} - SC_{it})^2}}{\sqrt{\frac{1}{T_0} \sum_{t=1}^{T_0} (Y_{it} - SC_{it})^2}} \quad (5)$$

where  $T_0$  is the number of pre-treatment periods and  $SC_{it}$  stands for the synthetic control. We also compute a normalised RMSPE for the whole post-treatment period as:

$$r_i = \frac{\sqrt{\frac{1}{T-T_0} \sum_{t=T_0+1}^T (Y_{it} - SC_{it})^2}}{\sqrt{\frac{1}{T_0} \sum_{t=1}^{T_0} (Y_{it} - SC_{it})^2}} \quad (6)$$

Then the p-value from the permutations for each post-treatment month is computed:

$$p_{it} = \frac{\sum_i I_+(r_{it} - r_{UK,t})}{N} \quad (7)$$

while for the whole post-treatment period we have:

$$p_i = \frac{\sum_i I_+(r_i - r_{UK})}{N} \quad (8)$$

where  $I_+(\cdot)$  returns one if the argument is non-negative and zero otherwise and 28 is the number of SC computed (UK plus the 27 EU members placebos). The p-values in (7) and (8) will be zero if the UK experienced the largest gap in post-treatment relative to pre-treatment fit and value of one if UK had the smallest gap.

Given that the application of the SC to international trade is not too widespread, a few considerations are in place. The nature of international trade is such changing trade costs with one partner affects trade with other partners as well. This creates a problem for the identification of trade policy parameters as the control group is also partly affected. However, we believe that the data presented in Figure 5-6 should provide reassurance that spill-over effects are unlikely to be large in this setting. Indeed, if spill-over effects are present they are likely to be larger for UK-ROW trade than for EU27-EU27 or EU-ROW trade, because the UK accounts for a small fraction of EU27 trade – hence small effects on EU27 multilateral resistance – while the EU accounts for large share of UK trade – hence larger effect on multilateral resistance of UK. However, we do not see marked differences between the trends in UK-ROW and EU-ROW trade flows, this suggests that spill-over effects are not a major issue in this setting. In terms of the SC, we have been careful in selecting the donor pool for a pair  $ij$  ( $i$ =exporter,  $j$ =importer) including exports of  $i$  to other countries and imports of  $j$  from other countries. This strategy is akin to the inclusion of exporter-time and importer-time fixed effects in the triple difference (or gravity) model, which are included to control for changes in the multilateral resistance terms. Finally, regarding Covid-19, our assumption is that the pandemic represents a country-specific shock. Hence, it would affect exports of country  $i$  to all destinations in the same manner, or imports of country  $j$  from all origins in a similar way. This again motivates the selection of the SC donor pool to include only within-exporter and within-importer flows.

In the Appendix, Section A.8 we also report an alternative inference method based on resampling to compare our results with those of Douch and Edwards (2021).

## 2.4 Covid-19

As Covid occurred, more or less, concurrently with the introduction of the TCA and had a substantial impact on trade, we need to ensure that we control for possible Covid confounding effects. We do so in several ways.

First, the evolution of the pandemic was different across countries in terms of timing and intensity. In order to control for this in the DD estimation we add in equations (1) and (2) either the (log of) number of Covid-19 cases per capita or the (log of) number of Covid-19 deaths per capita:

$$Y_{jt} = \exp[\alpha_j + \alpha_t + \beta(EU_j \times Post_t) + \gamma \ln Covid19_{jt}] + \epsilon_{jt} \quad (9)$$

Since both the number of cases and deaths take value of zero in some periods, we approximate the log with the inverse of the hyperbolic sine transformation  $\ln \left[ x + (x^2 + 1)^{1/2} \right]$ .

Second, in the triple difference approach we include reporter-time and partner-time dummies, in order to control *inter alia* for the different impact of Covid over time across countries. Similarly, in the SCM approach, our assumption is that the pandemic represents a country-specific shock. Hence, it would affect exports of country  $i$  to all destinations in the same manner, or imports of country  $j$  from all origins in a similar way. This again motivates the selection of the SC donor pool to include only within-exporter and within-importer flows.

Third, we need to consider the possibility that Covid may have had heterogeneous effects across products and therefore that our aggregate results are being affected by compositional effects at a more disaggregated level. We explore this possibility in two ways. First, we do so by removing the Covid-related products from our regressions – where we find this makes very little difference to the results. Second (see Appendix, Table 11) we estimate the model on more disaggregated data (at the HS 2-digit level), and where we control for reporter-product-year, partner-product-year and reporter-partner-product fixed effects. Once again we find that our results are highly robust.

## 3 Data

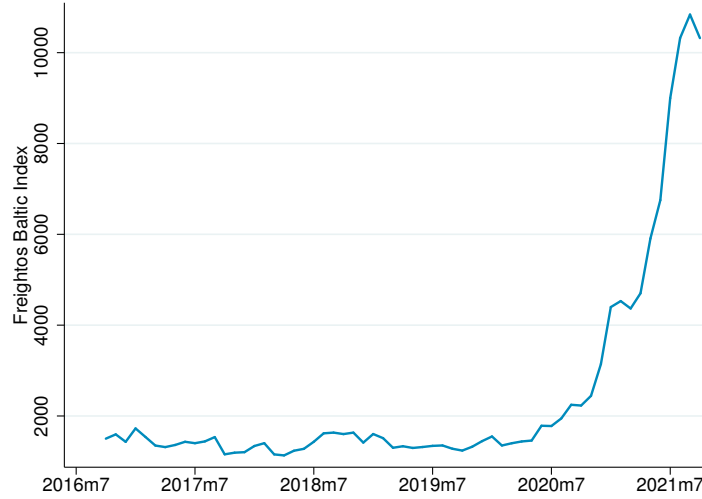
Trade data for the UK are downloaded from the HMRC while data for other EU countries are downloaded from Eurostat (COMEXT). For intra-EU trade, exports are preferred to imports because of Intrastat threshold method. Therefore intra-EU imports are replaced with their mirror flows. For the UK we use HMRC reported data for both exports and imports because of smaller changes in collection method in 2021 compared to Eurostat mirror flows. Differences between HMRC and Eurostat mirror flows are analysed in Gasiorek and Tamberi (2021). The main difference concerns UK exports to the EU as reported by the HMRC and its mirror flows (EU imports reported in Eurostat). Gasiorek and Tamberi (2021) conclude that the main driver of these differences the change in Eurostat collection method for trade data with the UK which passed from the Intrastat survey (the survey used to record intra-EU trade) to customs declaration. This change implied that EU imports from the UK passed from being recorded on a country of consignment basis to a country of origin basis. On the other hand, the same methodological change was not implemented by HMRC. Hence, HMRC reported data should be utilised to analyse UK-EU trade and TCA effects. Note moreover that HMRC started to record UK imports from the EU on a country of origin basis as opposed to country of consignment in 2022. This means that UK imports from the EU data have a structural break in 2022. We therefore limit our analysis of the TCA to the first year of implementation ending in December 2021.

When we look at aggregate trade we exclude gold trade (HS 7108). We also look at trade disaggregated by HS sections (21 sections) to examine whether the effect has been heterogeneous across product groups.

The dataset has the UK and the 27 EU members as reporters, and UK, EU27 and OECD + BRICS as partners. We exclude Japan and Canada because the EU and UK implemented FTAs with these countries in the period considered. For the analysis of the referendum, the dataset covers the period Jan2012-Dec2019, and we exclude Croatia both as exporter and importer as it joined the EU in 2013. For the analysis of the TCA, the period considered goes from July 2016 to December 2021.

In terms of other data sources, distance comes from CEPII<sup>12</sup>. For the cost of freight we use the Freight Baltic Index (FBX) computed by Freightos.<sup>13</sup>

Figure 4: Freight cost indexes, Freightos Baltic Index Global



Source: authors' elaboration of Freightos Baltic Index data.

Data on Covid-19 cases and deaths are downloaded from Our World in Data (OWID). We supplement OWID data with data from the European Centre for Disease Prevention and Control for China in December 2019. The raw data are in weekly frequency and are aggregated at monthly frequency.

## 4 Results

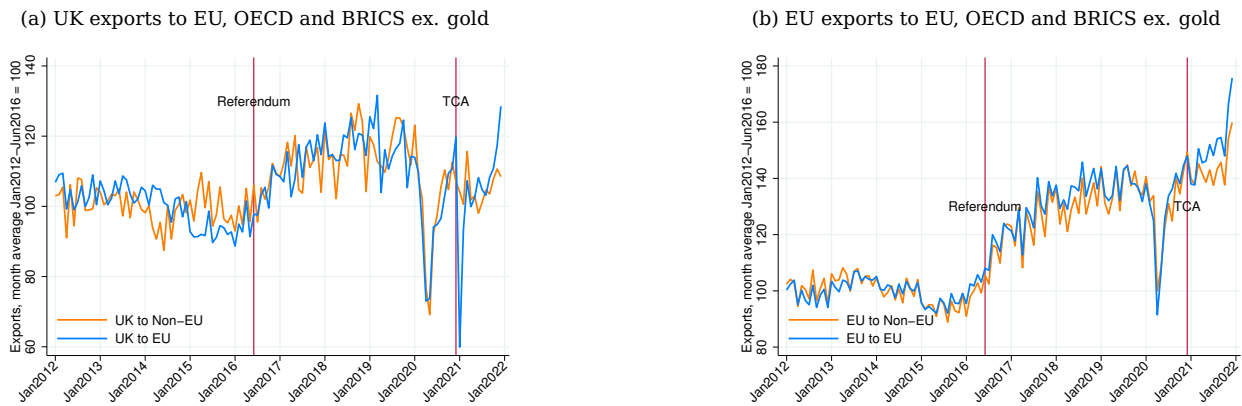
As preliminary evidence we plot the data used for the DD and DDD estimations. Figure 5 plots UK (panel a) and EU (panel b) export indices to the EU and the control group over the period January 2012-December 2021. Each point is normalised by the month average over the period January 2012-June 2016 – that is, the value for January 2018 is divided by the average of January 2012-16. These are essentially the data used in DD and DDD with normalisation accounted for by fixed effects. Figure 6 undertakes the same exercise for UK and EU imports. UK exports to the EU show a big drop in January 2021 only to realign with exports to the control group in the following months. On the other hand, EU exports to the EU (figure 5 panel b) do not

<sup>12</sup>From the CEPII GeoDist database we use the variable 'distw'.

<sup>13</sup>See <https://fbx.freightos.com/>

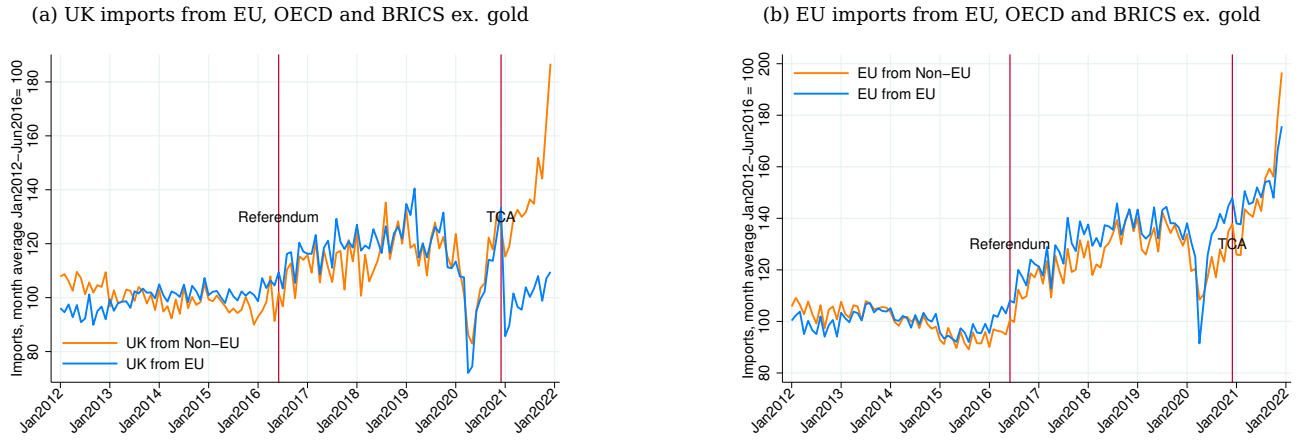
show such a sharp drop in January 2021. Note moreover how exports to the EU and control group move very closely for both the UK and the EU27 also during the pandemic period. This indicates that the group of OECD + BRICS countries represents a very good control for the EU. Turning to UK imports in Figure 6 (panel a) we see that imports from the EU and control group are very close prior to 2021, but as the TCA is introduced the two flow are separated by a gap of about 20 index-points. Nothing similar happens to EU imports from the EU. Moreover, we see that UK imports from the control group behave very similarly to EU imports from the control and from the EU, and the only different flow is UK imports from the EU.

Figure 5: UK and EU exports to EU and control group



Source: authors' elaboration of HRMC and Eurostat data. UK trade data are taken from HMRC trade statistics, while trade data of EU member states come from the Eurostat Comext database. The series are normalised by the month-average for the period Jan2012-Jun2016.

Figure 6: UK and EU imports from EU and control group



Source: authors' elaboration of HRMC and Eurostat data. UK trade data are taken from HMRC trade statistics, while trade data of EU member states come from the Eurostat Comext database. The series are normalised by the month-average for the period Jan2012-Jun2016.

#### 4.1 The Referendum effect

Table 1 reports the difference-in-differences and triple differences results for the 2016 referendum. The period used for the estimation is January 2012 to December 2019 with monthly data, and the post-referendum period is July 2016-December 2019. In this way we consider the referendum period excluding 2020, when Covid-19 hit. The sample excludes Croatia which joined the EU in 2013. The regressions yield small coefficients which are not statistically different from zero.



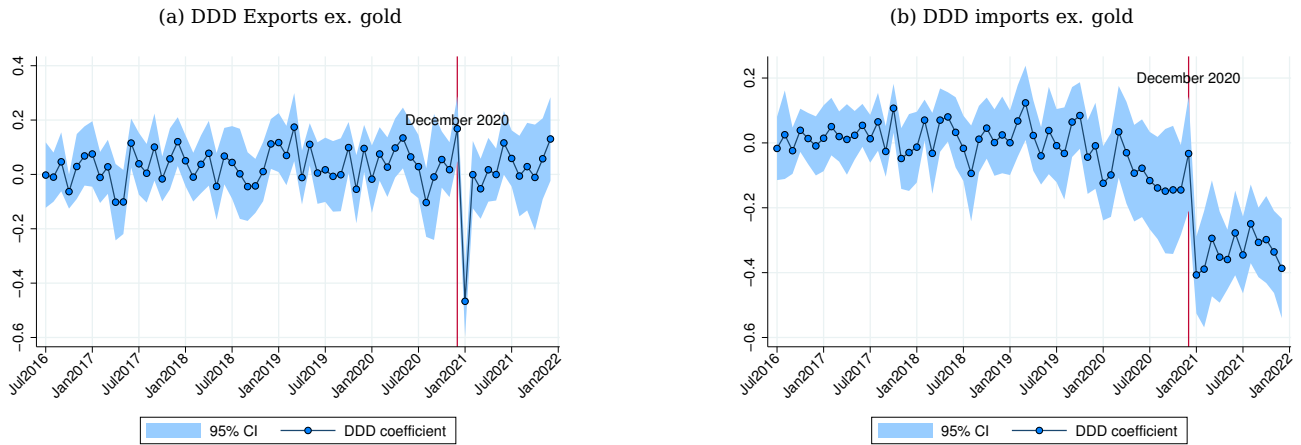
Table 1: Referendum results for total trade ex. gold, 2012-19

|                       | Exports          |                  | Imports          |                  |
|-----------------------|------------------|------------------|------------------|------------------|
|                       | (1)              | (2)              | (3)              | (4)              |
|                       | m1               | m2               | m3               | m4               |
| EU×Post-referendum    | 0.012<br>(0.047) |                  | 0.034<br>(0.053) |                  |
| UK×EU×Post-referendum |                  | 0.032<br>(0.033) |                  | 0.019<br>(0.034) |
| Pair FE               | Yes              | Yes              | Yes              | Yes              |
| Time FE               | Yes              | No               | Yes              | No               |
| Partner-time FE       | No               | Yes              | No               | Yes              |
| Reporter-time FE      | No               | Yes              | No               | Yes              |
| N                     | 4,128            | 108,960          | 4,128            | 108,960          |

Robust standard errors clustered at the partner level in parenthesis. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ . The table reports the DD and DDD estimates for the 2016 referendum effect on UK exports to and imports from the EU. The DD sample includes UK trade with the EU (ex. Croatia), OECD (ex. Canada and Japan) and BRICS countries. The DDD sample also includes EU (ex. Croatia) trade with the EU, OECD (ex. Canada and Japan) and BRICS countries.

We also estimate the triple-difference equation over the period 2012-21 interacting the UK-EU indicator with the post-referendum dummy (taking the value of one from July 2016 to December 2021) with time dummies, including reporter-time, partner-time and reporter-partner-month fixed effects. This allows us to see the evolution of the treatment effect relative to the pre-referendum period Jan2012-Jun2016. The estimated coefficients with the 95% confidence interval are plotted in Figure 7 for exports (panel a) and imports (panel b). For exports, we find no sign of a treatment effect in any of the post-referendum periods, but with a large drop in January 2021. Similarly, none of the imports regression coefficients in panel b are statistically significant before 2021, and it is only when the TCA enters into force in 2021 that UK imports from the EU drop.

Figure 7: UK and EU imports from EU and control group



The figure reports triple difference coefficients for UK-EU trade. The pre-treatment period is Jan2012-Jun2016. The PPML regressions include reporter-partner-month, reporter-time and partner-time fixed effects. Standard errors are clustered by reporter-partner.

We then run the Synthetic Control matching UK-EU trade over the period Jan2012-Jun2016. This technique shows no effect on UK exports to the EU in the post-referendum period, apart for the sharp drop which occurred in January 2021. For imports, the SC shows that UK imports from the EU started to decrease in the second half of 2019, and then felt substantially in 2021. Results are reported in Figures 11-12 in the Appendix.

Our results are in contrast with those of Douch and Edwards (2021) who find strong negative effects of the 2016 referendum on UK trade applying the Synthetic Control, both with EU and non-EU countries. The authors also find that UK trade started to decrease in 2015. There are some differences between their approach and ours. First, Douch and Edwards (2021) focus on the EU14 rather than the EU27. Second, they use data from IMF which are on a Balance of Payment basis, so coverage is slightly different from ours which is on an IMTS basis. Their flows are seasonally adjusted while ours are not, and their data are measured in US dollars while ours are in pound sterling. Third, they exclude any trade flow involving the UK or other EU countries from the donor pool. Under the assumption that UK trade with rest of the world (ROW) is also affected by Brexit, Douch and Edwards (2021) estimate treatment effects for UK-ROW trade. While the baseline specification shows that UK exports to the EU are more affected than UK exports to ROW, robustness such as inclusion of EU flows in donor pool, changing the sample size or including the whole EU27 yields treatment effects larger for UK-ROW than UK-EU trade.<sup>14</sup> The fact that effects are not systematically larger for UK-EU flows than UK-ROW flows raises the question whether the effects picked up by Douch and Edwards (2021) are really driven by the Brexit referendum. While it is certainly possible that Brexit impacted UK trade beyond the EU, it is harder to believe that effects would be larger for non-EU trade than for EU trade.

Finally, the method used by Douch and Edwards (2021) to make inference is different from ours and to that which is standard in the SC literature (see Abadie, Diamond and Hainmueller (2010) and Abadie (2021)).

<sup>14</sup>The authors estimate treatment effects for three periods: Jun2015-Jun2016, Jul2016-Feb2017 and Mar2017-Mar2018. The baseline results for UK exports to the EU14 in the three periods are -20.8%, -25% and -20.3%, while for UK exports to ROW are -3.6%, -13.4%, -15.2%. Including EU flows, the results for UK exports to the EU are -5.4%, -9.8% and -10.9%. See also tables B9-B13 in the appendix of Douch and Edwards (2021).

Rather, Douch and Edwards (2021) use the method of Saia (2017) which is based on re-sampling the donor pool to construct a standard error of the treatment effect. However, this is a different measurement of uncertainty regarding the estimate compared to the permutation method of Abadie, Diamond and Hainmueller (2010), and more similar to the leave-one-out test also proposed by Abadie, Diamond and Hainmueller (2010). Importantly, this method does not tell us whether the SC applied to other countries would yield similar treatment effects. For robustness we have also replicated this inference exercise and report the results in Section A.8 in the Appendix. While we confine detailed discussion in the Appendix, our results do not point to particularly significant post-referendum effects even when using the resampling method for inference.

On the other hand, our results are in line with the conclusions of Dhingra and Sampson (2022) who note that, while studies such as Graziano, Handley and Limão (2021) and Crowley, Exton and Han (2018) provide evidence that uncertainty had some effects on trade flows, ‘the magnitude of these responses was insufficient to cause a noticeable shift in the geography of aggregate UK trade’. Moreover, in reviewing the ‘Brexit literature’ Dhingra and Sampson (2022) conclude in line with Ayele and Winters (2020) that there is no evidence that the exchange depreciation occurred in the aftermath of the 2016 referendum boosted British exports.

## 4.2 The TCA effect

The previous section showed that the 2016 referendum had little to no impact on aggregate UK-EU trade in the years following the referendum. While we do not exclude the possibility that adjustments to uncertainty about future trade costs were taking place in the post-referendum period, using a range of econometric techniques we could not find evidence of a negative impact of the referendum on aggregate UK-EU trade. Hence, we now focus on the post-referendum period to estimate more precisely the impact of the TCA on UK-EU trade. Because we do not exclude the possibility of structural adjustments taking place after the referendum, and because as observed in Figures 5-6 both UK and EU trade started to grow substantially in 2016, we restrict our analysis to the post-referendum period 2017-2021.

## 4.3 DD total trade

Results for the DD estimation are reported in Table 2. When we control only for time and partner FE in Table 2, we get a small negative effect on exports (-1.5% over the whole 2021) which is not significant at conventional levels of significance. Results by month tell us that there was a substantial drop in January 2021 (-41%) followed by a quick recovery. On the other hand, UK imports have been consistently affected in all the months of 2021 and over the whole year they were down by 32%. Coefficient estimates are translated into percentage effects by computing  $\exp(b)-1$ .

We find no difference in results when we control for Covid-19 in Table 12, either when we control for Covid-19 cases or deaths. On the other hand, results become very different when we control for the cost of freight costs in Tables 13-14. In Table 13 we control for the log of freight cost interacted with distance. In Table 13 we also add the log of Covid-19 deaths as a control. Finally, in Table 14 we use the interaction of distance with time-dummies to account for changes in freight costs and we also add Covid-19 deaths as control in the last two columns. For UK exports to the EU, we now get larger negative and significant effects, with a cumulative decline over 2021 of 22-23%. For imports, the results remain negative and significant in almost all months of 2021 although reduced in magnitude. However, these strong negative and significant results for UK exports to the EU are not present when we use DDD or SC.

Table 2: DD results total trade ex. gold

|                  | Exports           | Exports              | Imports              | Imports              |
|------------------|-------------------|----------------------|----------------------|----------------------|
| EU×2021          | -0.015<br>(0.057) |                      | -0.383***<br>(0.084) |                      |
| EU×Jan×2021      |                   | -0.533***<br>(0.082) |                      | -0.304***<br>(0.111) |
| EU×Feb×2021      |                   | -0.076<br>(0.067)    |                      | -0.390***<br>(0.150) |
| EU×Mar×2021      |                   | -0.089<br>(0.058)    |                      | -0.340*<br>(0.176)   |
| EU×Apr×2021      |                   | 0.002<br>(0.049)     |                      | -0.358***<br>(0.113) |
| EU×May×2021      |                   | -0.001<br>(0.049)    |                      | -0.357***<br>(0.064) |
| EU×Jun×2021      |                   | 0.087<br>(0.055)     |                      | -0.314***<br>(0.069) |
| EU×Jul×2021      |                   | 0.046<br>(0.059)     |                      | -0.317***<br>(0.080) |
| EU×Aug×2021      |                   | 0.022<br>(0.076)     |                      | -0.288***<br>(0.093) |
| EU×Sep×2021      |                   | 0.059<br>(0.084)     |                      | -0.389***<br>(0.088) |
| EU×Oct×2021      |                   | 0.009<br>(0.105)     |                      | -0.448***<br>(0.149) |
| EU×Nov×2021      |                   | 0.092<br>(0.093)     |                      | -0.431***<br>(0.139) |
| EU×Dec×2021      |                   | 0.076<br>(0.075)     |                      | -0.596***<br>(0.133) |
| Partner FE       | Yes               | No                   | Yes                  | No                   |
| Time FE          | Yes               | Yes                  | Yes                  | Yes                  |
| Partner-month FE | No                | Yes                  | No                   | Yes                  |
| Observations     | 2640              | 2640                 | 2640                 | 2640                 |

Robust standard errors clustered at the partner level in parenthesis. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The table reports the DD estimates for UK exports to and imports from the EU.

#### 4.4 DDD total trade

The DDD estimation show a non-significant -0.036 coefficient on UK exports to the EU over the period January-December 2021, with a sharp drop in January 2021 (-40%, strong significance). The result for the specification of Table 3 is robust to controlling for shipping costs, either interacting distance with freight cost indices or with time dummies (Table 15). As in the DD estimation, imports appear to have been affected in all the months of 2021 and the results are robust to controlling for freight costs. These results, differently from the

DD specification, are robust to any shock affecting a given country in a particular time period. We therefore consider these more reliable than those of the DD specification. In Table 16 in the Appendix we also report a robustness test where we exclude the FTA partners from the control group, and we find very similar results.

Table 3: DDD results total trade ex. gold

|                  | Exports           | Exports              | Imports              | Imports              |
|------------------|-------------------|----------------------|----------------------|----------------------|
| UK×EU×2021       | -0.036<br>(0.049) |                      | -0.323***<br>(0.044) |                      |
| UK×EU×Jan×2021   |                   | -0.520***<br>(0.065) |                      | -0.381***<br>(0.060) |
| UK×EU×Feb×2021   |                   | -0.033<br>(0.064)    |                      | -0.412***<br>(0.080) |
| UK×EU×Mar×2021   |                   | -0.118***<br>(0.045) |                      | -0.330***<br>(0.093) |
| UK×EU×Apr×2021   |                   | 0.009<br>(0.054)     |                      | -0.374***<br>(0.065) |
| UK×EU×May×2021   |                   | -0.013<br>(0.045)    |                      | -0.359***<br>(0.050) |
| UK×EU×Jun×2021   |                   | 0.055<br>(0.049)     |                      | -0.292***<br>(0.055) |
| UK×EU×Jul×2021   |                   | 0.024<br>(0.054)     |                      | -0.314***<br>(0.060) |
| UK×EU×Aug×2021   |                   | 0.017<br>(0.070)     |                      | -0.200***<br>(0.062) |
| UK×EU×Sep×2021   |                   | 0.016<br>(0.072)     |                      | -0.277***<br>(0.050) |
| UK×EU×Oct×2021   |                   | -0.036<br>(0.095)    |                      | -0.317***<br>(0.082) |
| UK×EU×Nov×2021   |                   | 0.048<br>(0.084)     |                      | -0.268***<br>(0.059) |
| UK×EU×Dec×2021   |                   | 0.003<br>(0.066)     |                      | -0.370***<br>(0.075) |
| Partner-time FE  | Yes               | Yes                  | Yes                  | Yes                  |
| Reporter-time FE | Yes               | Yes                  | Yes                  | Yes                  |
| Pair FE          | Yes               | No                   | Yes                  | No                   |
| Pair-month FE    | No                | Yes                  | No                   | Yes                  |
| Observations     | 72300             | 72300                | 72300                | 72300                |

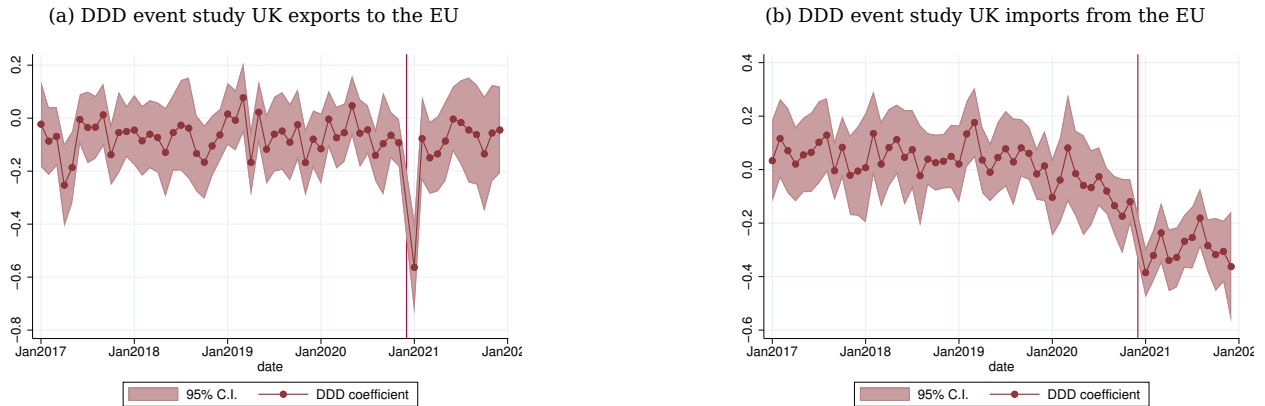
Robust standard errors clustered at the exporter-importer level in parenthesis. \* p < 0.10, \*\* p < 0.05, \*\*\* p < 0.01. The table reports the DDD estimates for UK exports to and imports from the EU.

A possible concern with the estimation presented above is whether trade in Covid-related products are driving our results. We therefore select the Covid-related products as defined in the Eurostat Comext database

and remove them from total trade.<sup>15</sup> The estimation results for this exercise are reported in Table 17 in the Appendix, and they confirm the baseline results. UK exports appear affected only in January and March 2021, with the overall effect in 2021 being -0.05 not statistically significant. On the other hand, for imports we still find strong and persistent effects in all months of 2021.

The estimation of Table 3 does not deal explicitly with the possibility of anticipation effects affecting trade flows before 2021. However, the results reported in Figure 7 looking at a longer pre-treatment effect do not show strong sign of a referendum effect nor of anticipation of TCA. These results are in line with the simulations of Steinberg (2019) which finds that referendum-induced uncertainty had little impact on UK macroeconomic variables. At the same time, we do not exclude the possibility of anticipation effects occurring in the months leading to the TCA. To see whether the data show any sign of an anticipation effect, we estimate the triple-difference equations using an event study approach. This is done by interacting the EU dummy with dummies for each month, using as the reference period December 2020 – the last pre-treatment period. For this event study approach we use all the months over the period January 2017 to December 2021. The results are reported in Figure 8 where we plot the estimated coefficients together with the 95% confidence interval based on robust s.e. clustered at the pair level. The figure for exports (panel a) does not show any sign of anticipation and we can see the sharp drop in January 2021. The decline in imports on the other hand seems to have started around July-August 2020, only to drop even further in January 2021. Note however that these regressions do not account for seasonality and are therefore less accurate than the results reported in Table 3 or those of Figure 7 where the baseline period is Jan2012-Jun2016.

Figure 8: DDD event study coefficients



Confidence interval based on robust standard errors clustered at the exporter-importer level in parenthesis. The figures report the DDD event-study estimates for UK exports to and imports from the EU. The reference period is December 2021.

## 4.5 DDD HS sections

In this section we report on the results of the triple difference estimation by HS section (21 sections). We estimated equation (3) and (4) for each section separately. The results for exports (Table 4) are more heterogeneous than those for imports (Table 5). For exports, the products which have been negatively affected

<sup>15</sup>The list of products is available at <https://ec.europa.eu/eurostat/documents/6842948/11003521/Corona+related+products+by+categories.pdf>

over the period January-December 2021 are Vegetables (-34%), Fats & oils (-57%), Food, beverages & tobacco (-15%), Plastic and rubber (-11%), Textile (-59%), Footwear (-72%), Ceramic and glass (-15%), Gold and pearls (-38%), and Miscellaneous manufacturing (-21%). Most sectors experienced a sharp drop in January 2021, but not in the other months of 2021. Note that the largest export sectors – Chemicals, Machineries and Transport equipment – did not see a significant drop in 2021, in line with the results for total exports. On the other hand, we see that for imports the effect over the months of January-December 2021 are not significant for only five of the twenty-one HS sections.

Table 4: DDD results exports by HS sections

| HS section                  | Jan-Dec   | Jan       | Feb       | Mar       | Apr       | May       | Jun       | Jul       | Aug       | Sep       | Oct       | Nov       | Dec       |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Animals & animal products   | -0.037    | -0.355    | -0.152    | 0.150     | 0.278     | 0.116     | 0.014     | -0.197    | -0.102    | -0.012    | 0.022     | -0.139    | -0.179    |
| Vegetable products          | -0.423*** | -1.003*** | -0.497**  | -0.398*   | -0.608**  | -0.337**  | -0.616*** | -0.662*** | -0.333**  | -0.186    | -0.695*** | -0.082    | 0.062     |
| Animal and veg. fats & oils | -0.835*** | -0.332    | -1.241**  | -0.204    | -1.472*** | -1.165*** | -1.413**  | -0.312    | -0.676*   | -0.816*** | 0.102     | 0.418     | -0.708*   |
| Foodstuffs, bev. & tobacco  | -0.166**  | -1.047*** | -0.381*** | -0.196    | -0.112    | -0.019    | -0.172    | -0.037    | -0.026    | -0.134*   | -0.190**  | -0.085    | 0.002     |
| Mineral products            | 0.201     | -0.050    | 0.339     | 0.389     | 0.348     | 0.466     | 0.937*    | -0.559*   | 0.208     | 0.583     | 0.298     | 0.711     | -0.562*   |
| Chemicals                   | -0.029    | -0.592*** | 0.090     | -0.137    | -0.050    | -0.115    | 0.160     | 0.000     | 0.149     | -0.120    | -0.148    | 0.070     | 0.179     |
| Plastics & rubber           | -0.121*   | -0.588*** | -0.102    | -0.092    | -0.095    | -0.054    | 0.023     | -0.105    | -0.119    | -0.061    | -0.169**  | -0.145    | -0.058    |
| Leather                     | -0.279    | -0.653    | -0.275    | -0.080    | -0.061    | -0.416*   | -0.267    | -0.296    | -0.082    | -0.243    | -0.210    | -0.328**  | -0.488*** |
| Wood                        | 0.060     | -0.747*** | 0.251     | 0.144     | 0.612***  | 0.482     | 0.215     | 0.152     | -1.055**  | 0.317     | 0.168     | 0.467*    | 0.340     |
| Paper                       | -0.070    | -0.537*** | 0.044     | 0.053     | -0.084    | -0.055    | -0.010    | -0.194    | -0.049    | -0.078    | -0.013    | 0.028     | -0.052    |
| Textile & clothing          | -0.886*** | -1.472*** | -1.032*** | -0.665*** | -0.928*** | -0.992*** | -0.952*** | -0.864*** | -0.764*** | -0.720*** | -0.657*** | -0.735*** | -1.084*** |
| Footwear & headgear         | -1.282*** | -2.276*** | -1.692*** | -1.332*** | -1.324*** | -1.455*** | -1.306*** | -1.448*** | -0.689**  | -0.772*** | -0.878*** | -1.160*** | -1.673*** |
| Ceramic & glass             | -0.162**  | -0.667*** | -0.386**  | -0.166    | -0.092    | -0.131    | -0.134*   | 0.013     | -0.018    | -0.118*   | -0.193**  | -0.148*   | -0.040    |
| Gold & pearls               | -0.474*   | -0.554*   | -1.556*** | -1.526*** | -0.078    | 1.016***  | 0.532**   | 0.614**   | 0.806***  | -1.138**  | -1.380*** | -0.727**  | -0.530    |
| Metals                      | -0.076    | -0.617*** | -0.015    | -0.246*** | 0.057     | -0.135*   | -0.105    | -0.107    | -0.035    | 0.041     | 0.044     | -0.024    | 0.064     |
| Machinery & electrical eq.  | 0.050     | -0.524*** | 0.005     | -0.001    | 0.135*    | 0.073     | 0.160**   | 0.148*    | 0.115*    | 0.171**   | -0.048    | 0.059     | 0.135*    |
| Transport eq.               | -0.067    | -0.462*** | 0.104     | -0.295**  | -0.061    | -0.244**  | -0.083    | 0.234*    | 0.004     | -0.046    | -0.025    | 0.141*    | -0.080    |
| Precision tools             | 0.122**   | -0.316*** | 0.146*    | 0.229***  | 0.221**   | 0.182*    | 0.082     | 0.159**   | 0.089     | 0.123*    | 0.010     | 0.206***  | 0.192***  |
| Arms & ammunitions          | -0.113    | -0.470    | -1.200**  | 0.170     | 0.253     | 0.422     | -0.355    | 0.713***  | 0.390     | 0.241     | -0.609**  | -0.669**  | -0.659*** |
| Miscellaneous manuf.        | -0.232**  | -0.804*** | -0.089    | -0.211*   | -0.086    | 0.100     | -0.085    | -0.332**  | -0.245**  | -0.413*** | -0.217    | -0.281**  | -0.308*** |
| Art & antiques              | 2.169***  | 1.067*    | 2.804***  | 1.514***  | 2.501***  | 3.305***  | 1.835***  | 2.214***  | 2.382***  | 2.148***  | 1.776***  | 2.480***  | 2.018***  |

The table reports the DDD coefficients for each HS section including exporter-time, partner-time and exporter-importer fixed effects. For the month-specific effects, the regressions include importer-exporter-month fixed effects. \* p < 0.10, \*\* p < 0.05, \*\*\* p < 0.01.

Table 5: DDD results imports by HS sections

| HS section                  | Jan-Dec   | Jan       | Feb       | Mar       | Apr       | May       | Jun       | Jul       | Aug       | Sep       | Oct       | Nov       | Dec       |
|-----------------------------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| Animals & animal products   | -0.558*** | -0.533*** | -0.655*** | -0.592*** | -0.570*** | -0.590*** | -0.463**  | -0.631*** | -0.618*** | -0.459*   | -0.650*** | -0.464**  | -0.557**  |
| Vegetable products          | -0.252*** | -0.202    | -0.335**  | -0.280**  | -0.480*** | -0.286*** | -0.183**  | -0.198    | -0.067    | -0.161    | -0.313*** | -0.164    | -0.454*** |
| Animal and veg. fats & oils | -0.496*   | -0.101    | -1.504**  | -0.236    | -0.797**  | -0.841*** | -0.616    | -0.455*   | -0.220    | -0.083    | -0.543    | -0.262    | -0.325    |
| Foodstuffs, bev. & tobacco  | -0.201*** | -0.119    | -0.236*   | -0.224**  | -0.222**  | -0.159    | -0.158    | -0.211*   | -0.205**  | -0.179*   | -0.229*   | -0.182**  | -0.270**  |
| Mineral products            | -0.241**  | -0.217    | -0.117    | -0.306    | -0.288    | -0.410*   | -0.187    | -0.076    | 0.043     | -0.090    | -0.249    | -0.200    | -0.482    |
| Chemicals                   | -0.563*** | -0.617**  | -1.020*** | -0.793*** | -0.553*** | -0.765*** | -0.505**  | -0.541*** | -0.462*** | -0.366*** | -0.561*** | -0.406**  | -0.227    |
| Plastics & rubber           | -0.150*** | -0.272*** | -0.329*** | -0.338*** | -0.288*** | -0.194*** | -0.053    | -0.142    | -0.040    | 0.003     | -0.107    | -0.051    | -0.096*   |
| Leather                     | -0.415*** | -0.730*** | -0.652*** | -0.534*** | -0.547**  | -0.414*   | -0.433**  | -0.505*** | -0.455*** | -0.261*   | -0.358*** | -0.266*   | -0.106    |
| Wood                        | -0.005    | -0.179    | -0.086    | 0.018     | 0.170     | 0.037     | 0.033     | -0.144    | 0.117     | 0.082     | -0.102    | 0.025     | -0.106    |
| Paper                       | -0.060    | -0.162*   | -0.023    | -0.186*** | -0.139*   | -0.129    | 0.090     | -0.084    | 0.041     | 0.024     | -0.079    | -0.017    | -0.133    |
| Textile & clothing          | -0.134**  | -0.309*** | -0.332*** | -0.207*** | -0.244*** | -0.164**  | -0.141*   | -0.282*** | -0.073    | 0.096     | 0.129     | -0.108    | -0.213**  |
| Footwear & headgear         | -0.573*** | -0.427*** | -0.786*** | -0.573*** | -0.357**  | -0.593*** | -0.639*** | -0.757*** | -0.451**  | -0.556**  | -0.463**  | -0.640*** | -0.668*** |
| Ceramic & glass             | -0.265*** | -0.332*** | -0.514*** | -0.355*** | -0.275*** | -0.205*** | -0.219**  | -0.500*** | -0.179*** | -0.150**  | -0.149    | -0.157**  | -0.191*** |
| Gold & pearls               | -0.329    | -0.711*** | -0.109    | -0.663**  | -0.605*   | -0.784*   | -1.693*** | 0.217     | 0.092     | 0.421     | 0.159     | -0.535*   | -0.363    |
| Metals                      | -0.344*** | -0.276*** | -0.240*** | -0.316*** | -0.270*** | -0.283*** | -0.472*** | -0.382*** | -0.348*** | -0.448*** | -0.325*** | -0.413*** | -0.336*** |
| Machinery & electrical eq.  | -0.201**  | -0.371*** | -0.302**  | -0.212*   | -0.290*** | -0.189**  | -0.114    | -0.153    | -0.155*   | -0.085    | -0.186**  | -0.114    | -0.295*** |
| Transport eq.               | -0.399*** | -0.248*   | -0.223**  | -0.069    | -0.290    | -0.505*** | -0.084    | -0.538**  | -0.080    | -0.740*** | -0.586*** | -0.748*** | -0.514*** |
| Precision tools             | -0.203**  | -0.166*   | -0.241*** | -0.107    | -0.211*   | -0.269**  | -0.275**  | -0.179    | -0.203**  | -0.167*   | -0.244**  | -0.134    | -0.258**  |
| Arms & ammunitions          | -0.512*** | -2.046*** | -0.244    | 0.021     | 0.057     | -0.017    | 0.161     | 0.025     | -1.139*** | -1.186*** | -0.861*** | -0.639    | 0.201     |
| Miscellaneous manuf.        | -0.251*** | -0.298*** | -0.425*** | -0.396*** | -0.257*** | -0.251*** | -0.327*** | -0.355*** | -0.259*** | -0.066    | -0.185*** | -0.038    | -0.276*** |
| Art & antiques              | -0.361*** | -0.127    | 0.861**   | -0.761**  | 1.173*    | 0.659*    | -3.222*** | 0.294     | -2.072*** | -2.117*** | -0.063    | 0.246     | -2.165*** |

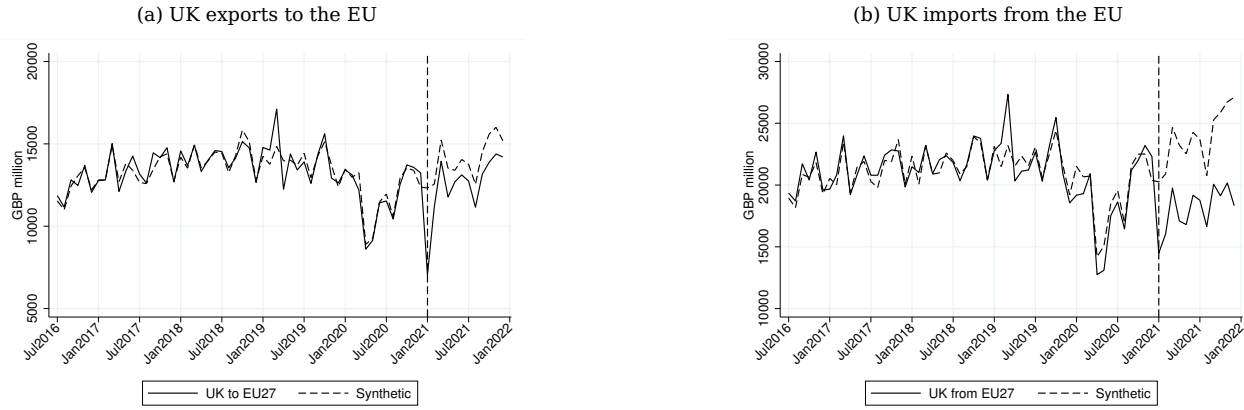
The table reports the DDD coefficients for each HS section including exporter-time, partner-time and exporter-importer fixed effects. For the month-specific effects, the regressions include importer-exporter-month fixed effects. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

## 4.6 Synthetic control

Figure 9 plots the outcome of the SC analysis for both exports (panel a) and imports (panel b). Calculation of p-values with the permutation method shows that for exports the UK was an extreme only in January 2021. Consistent with the other methodologies applied, for imports we find that the UK performed particularly poorly and permutation tests show significant effects in each month of 2021, apart August and September. For exports, over the period January-December 2021 we find a TCA effect of -11.85%. For imports, the cumulative effect is -24.05% and similar across the months. Table 6 reports the summary results for UK exports to and imports from the EU. Detailed results of the SC by partner and months are reported in separate tables in Tables 18-21 in the Appendix.



Figure 9: UK-EU trade SCM result



Synthetic control results for UK exports to and imports from the EU. The SC is estimated over the period Jul2016-Dec2020 for each EU member state and then aggregated to the EU27 total. For UK exports to (imports from) a given EU member state, the donor pool is composed by UK exports to (imports from) OECD and BRICS countries excluding Canada and Japan, as well the EU member state imports from (exports to) other EU members, OECD and BRICS countries excluding Canada and Japan.

Table 6: Synthetic control results

| Period       | Exports      |         | Imports      |         |
|--------------|--------------|---------|--------------|---------|
|              | % difference | p-value | % difference | p-value |
| Jan-Dec 2021 | -11.85       | 0.04    | -24.05       | 0       |
| Jan 2021     | -42.76       | 0       | -28.48       | 0       |
| Feb 2021     | -10.88       | 0.04    | -23.24       | 0       |
| Mar 2021     | -8.46        | 0.07    | -19.89       | 0       |
| Apr 2021     | -12.85       | 0.07    | -26.42       | 0       |
| May 2021     | -5.21        | 0.61    | -25.4        | 0       |
| Jun 2021     | -6.67        | 0.29    | -20.99       | 0       |
| Jul 2021     | -7.37        | 0.43    | -20.8        | 0       |
| Aug 2021     | -11.08       | 0.29    | -19.81       | 0.04    |
| Sep 2021     | -9.33        | 0.32    | -20.64       | 0.04    |
| Oct 2021     | -10.96       | 0.29    | -26.06       | 0       |
| Nov 2021     | -10.14       | 0.25    | -24.54       | 0       |
| Dec 2021     | -6.47        | 0.50    | -32.33       | 0       |

Note: *p*-values based on the normalised RMSPE.

In Figure 14 in the Appendix we report the results where instead of matching over all pre-treatment periods of the outcome variable, we match over the average of each months. That is, our predictors are the average of the outcome variable over January 2017-20, the average over February 2017-20 and so on which means that we have 12 predictors. Results are consistent with the ones presented above showing an overall decline

in exports of -13.9% and in imports of -27.1%, with similar patterns across months to the main specification.

Table 7 reports the top 20 donors for the baseline SC matched on all pre-treatment periods. As we estimated SC for UK trade with each EU27 member separately, the EU member weights have been aggregated to UK-EU27 level weighting each EU member by its share in UK trade over the period Jul2016-Dec2020. The pairs should be read as exporter-importer. For UK exports to the EU, the largest contributors are UK exports to Russia (9.5%), France to Germany (8.2%) and UK to South Africa (7.2%). Together, the top 20 donors account for 64% of the weights for UK exports to the EU. For UK imports from the EU the most important flow is UK imports from Switzerland (17.1%), followed by German exports to France (11.2%) and US exports to the UK (8.4%). The donor weights for imports are more concentrated than for exports, and together the top 20 donors account for 76% of total weights.

Note that while we report only the top 20 countries, in the estimation of the SC for each EU member state we find that weights are sparse, and that a large portion of donors attract zero weights. Considering that for trade of the UK with each EU member state the donor pool is composed of 60 control flows (17 flows for UK to ROW, 43 flows of EU member states trade with EU27 and ROW), for the EU27 total we have a donor pool of  $27 \times 60 = 1,620$  units. Across all SC estimation, the number of units which attract non-zero weights are 179 for exports and 154 for imports. However, many of these donors attract small weights, making the SC results sparse. As explained in Abadie (2021), this sparsity makes the SC results transparent and interpretable.

Table 7: Top 20 donors for Synthetic Control

| Exports to EU |        |       |        | Imports from EU |        |       |        |
|---------------|--------|-------|--------|-----------------|--------|-------|--------|
| Pair          | weight | Pair  | weight | Pair            | weight | Pair  | weight |
| GB-RU         | 0.095  | DE-FR | 0.020  | CH-GB           | 0.171  | FR-DE | 0.021  |
| FR-DE         | 0.082  | GB-CH | 0.018  | DE-FR           | 0.112  | KR-GB | 0.020  |
| GB-ZA         | 0.072  | IE-ES | 0.016  | US-GB           | 0.084  | CO-GB | 0.019  |
| RU-FR         | 0.061  | LT-PL | 0.015  | NO-GB           | 0.058  | BE-FR | 0.019  |
| NO-NL         | 0.039  | SE-HU | 0.013  | DE-US           | 0.043  | FR-CN | 0.015  |
| GB-KR         | 0.035  | GB-IS | 0.013  | NL-FR           | 0.038  | AT-NO | 0.015  |
| CH-DE         | 0.034  | GB-CN | 0.013  | IN-GB           | 0.027  | NL-DE | 0.014  |
| RU-DE         | 0.031  | LU-SE | 0.012  | ES-IT           | 0.025  | IT-IN | 0.014  |
| GB-US         | 0.027  | IE-CZ | 0.012  | FR-PT           | 0.025  | ZA-GB | 0.011  |
| NO-BE         | 0.022  | DE-BE | 0.012  | IT-ES           | 0.022  | PL-IT | 0.011  |

Another paper that applied the Synthetic Control to estimate the impact of the TCA on UK trade is Springford (2022). However, rather than looking at UK-EU trade, Springford (2022) estimates the SC on total UK trade with the world and separating out exports and imports. The author estimates that the TCA decreased total UK exports by 15.7% in the first year of the TCA. Two aspects are worth mentioning: as for the discussion of the referendum results of Douch and Edwards (2021), we believe that geography matters and that a credible estimate of TCA effects should show different impacts between UK-EU trade and UK-ROW trade. While it is possible that UK trade with ROW is also affected, one would expect the effect to be higher with the EU to be attributable to the TCA. Second, and perhaps most importantly, the SC method does not yield a confidence interval and inference should be performed with permutation methods, which are absent in Springford (2022). Indeed, the results for the treatment effects obtained by Springford (2022) are similar to ours, but after we draw inference based on permutation tests we find that the case of UK exports is not always extreme – apart

for January 2021.

## 4.7 The difference between the impacts on exports and imports

Our empirical results consistently show that the TCA had a strong negative effect on UK imports from the EU, but there is little or no evidence of an aggregate negative effect on UK exports to the EU. As discussed earlier this contrasts somewhat with the conclusions of many of the ex-ante studies on the ‘predicted’ impact of Brexit (eg. Dhingra et al. (2017); Steinberg (2019); Felbermayr, Gröschl and Steininger (2021)) where the effects are both considerably more symmetric, and which except for the Steinberg (2019) paper, found larger effects on UK exports than on UK imports. The only paper we are aware of which predicted a small impact on exports is that of Kee and Nicita (2017), who suggested UK exports to the EU would decline by a modest 2%. However, they did not consider UK imports from the EU so the issue of symmetry/asymmetry is not addressed.

Asymmetric effects could be driven by several factors. One key factor could be the role of the composition of trade. We explore this in some detail in Section 4.7.1 below, and show why this does not appear to be driving the results. Other factors could be possible asymmetries in trade barriers (be these tariff or non-tariff barriers), or asymmetries in demand or supply responses between the UK and the EU. The former is certainly possible and the work of Nardotto and Sequeira (2021) cited earlier is suggestive of demand shifts as possible explanatory factors. It seems highly unlikely that the results are driven by asymmetric trade barriers. This is for two reasons. First, the TCA is very much a symmetric trade agreement; and secondly with regard to border controls as well as customs declarations the UK delayed introducing a range of these at the start of the TCA whereas the EU did not. Hence, if anything this would suggest a larger impact on exports as opposed to imports – yet we find the converse.

It therefore seems likely that the asymmetric effects will be driven either by differential demand side responses, or supply side responses. With regard to differential supply side responses, we explore whether *prima facie* this could explain the results, in Section 4.7.2, by exploring the importance of relative market size.

### 4.7.1 Comparison at HS2 level

One possible explanation for the asymmetry of imports and exports results can be the difference in the product composition of the two flows. If UK’s exports are concentrated in products which saw less of an increase in trade barriers due to the TCA, while EU exports were concentrated on mainly TCA-affected products, we could find no TCA effects on UK exports but strong effects on UK imports. To see whether this is the case, we perform a simple decomposition. Total exports  $X_{ij}$  from country  $i$  to country  $j$  are the sum of  $i$ ’s exports to  $j$  of all products  $h$ :  $X_{ij} = \sum_h X_{ij,h}$ . Consider now the proportional change  $\hat{X}_{ij} = X'_{ij}/X_{ij}$  in total between a reference period value  $X_{ij}$  and a new value  $X'_{ij}$ :

$$\hat{X}_{ij} = \frac{X'_{ij}}{X_{ij}} = \frac{\sum_h X'_{ij,h}}{X_{ij}} = \sum_h w_{ij,h} \hat{X}_{ij,h}$$

where  $w_{ij,h} = X_{ij,h}/X_{ij}$  is the share of product  $h$  in  $i$ ’s exports to  $j$  at the base period. This simple decomposition shows that the change measured on total exports is a weighted average of product-level proportional changes.

Consider now the asymmetry between total UK exports and imports from the EU. Denote  $w_h^X$  and  $w_h^M$  the UK imports and exports share for product  $h$  and let  $\hat{X}_h^X$  and  $\hat{M}_h$  be the product-level changes in UK exports

to and imports from the EU, respectively. If the product-level changes are the same across all products, then effects on aggregate trade will be the same for exports and imports irrespective of differences in weights. On the other hand, if product-level effects  $\hat{X}_h^X$  and  $\hat{M}_h$  are heterogeneous, effects on aggregate exports and imports can differ. Differences can be driven by two factors: differences in the export and import weights  $w_h^X$  and  $w_h^M$  (composition) or differences in product-level changes between exports and imports.

While the results by HS sections in section 4.5 already show some heterogeneity of product-level effects, here we go a bit more in detail. To explore which of these two factors is at play in the context of the TCA, we estimated the triple difference regression for each HS 2-digit product (96 products) and flow (exports and imports), and retrieved information on base period imports and exports product weights for the pre-treatment period 2017-20.

Table 8 reports the summary statistics for the estimated coefficients as well as the HS2 product weights. The average import coefficient is about 40% larger than the average export coefficient. There is also less variation in the import than in the export coefficients as shown by the standard deviations. On the other hand, the product weights have very similar summary statistics.

Table 8: Summary of HS2 DDD estimates and weights

|                | Mean   | sd    | Median | Min    | Max   | Count |
|----------------|--------|-------|--------|--------|-------|-------|
| DDD imports    | -0.281 | 0.397 | -0.234 | -1.491 | 2.021 | 96    |
| DDD exports    | -0.202 | 0.469 | -0.165 | -1.318 | 2.127 | 96    |
| Weight imports | 0.010  | 0.024 | 0.003  | 0.000  | 0.178 | 96    |
| Weight exports | 0.010  | 0.023 | 0.002  | 0.000  | 0.122 | 96    |

The table reports summary statistics for the estimated DDD coefficients by each HS 2-digit product for exports and imports as well as 2017-20 product weights in total exports and imports.

We compute aggregate trade effects taking weighted averages of HS2 coefficients for both exports and imports. This yields a -0.32 for imports and -0.07 for exports, which are close to the effects estimated on total trade. This should not come as a surprise given the use of the PPML estimator. We discuss the estimation of aggregate treatment effects with product-level data and the PPML estimator in detail in the Appendix. We also invert weights, weighting imports coefficients with exports weights and *vice versa*. This gives a -0.31 aggregate imports effect and -0.12 aggregate exports effect. The asymmetry persists.

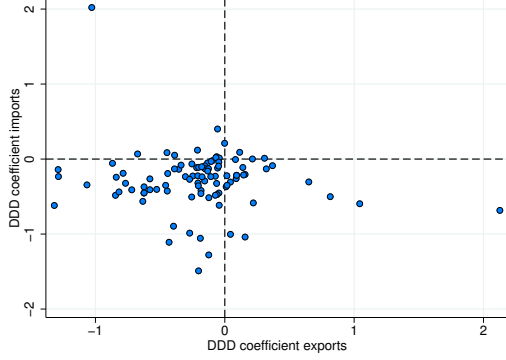
In figure 10 we compare the estimated DDD coefficients and the weights for exports and imports. In panel a we plot the imports against the exports DDD coefficients for each HS2 product. No clear pattern emerges from the scatter plot, and the correlation between the two coefficients is negative at -0.13. The HS2 coefficients identify two outliers: product 24 for UK imports (Tobacco) and product 97 for UK exports (Art and Antiques). If these two products are removed, the correlation between the imports and exports coefficients becomes positive but small at 0.043. However, these two products have very small shares – 0.002 and 0.004 for imports and 0.0001 and 0.0026 for exports, respectively – hence they do not drive aggregate results.

In panel b of Figure 10 we plot the UK imports weights against the UK exports weights computed over the period 2017-20 (which is the pre-treatment period for the coefficients of panel a). The import and export weights are very similar across HS2 products with a correlation of 0.85.

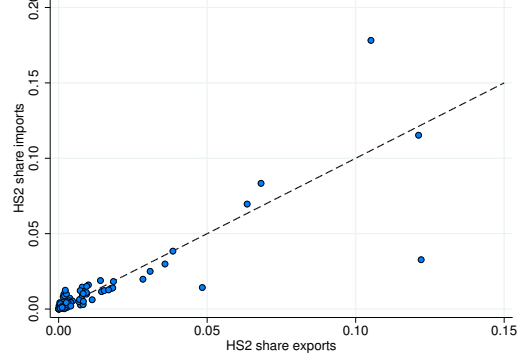
Overall, the decomposition exercise suggests that product composition of exports and imports does not explain the asymmetric TCA effects observed for the two flows.

Figure 10: TCA effects and pre-TCA trade weights by HS2

(a) Exports and imports DDD coefficients by HS2



(b) Exports and imports share by HS2



Panel (a) plots exports and imports triple difference coefficients for each HS 2-digit product. Panel (b) plots exports and imports weights in UK trade with the EU.

#### 4.7.2 Export market share

Another possible explanation for the asymmetry between exports and imports is the relative importance of the EU market for UK firms, in comparison to the importance of the UK market for EU firms. Over the period 2017-19, the UK exported 46.7% of its goods exports to the EU27. On the other hand, only 6.3% of total EU27 goods exports went to the UK over the same period. Hence, for UK firms dismantling trade relations with the EU may be very costly, while ceasing to serve the UK market is likely to be relatively less important for EU exporters. While we acknowledge that there could be other (not necessarily alternative) explanations to the difference between exports and imports results, measuring other potential drivers precisely can be hard and goes beyond the scope of this paper. In this section we test whether data provides support for the market size hypothesis as an explanation to the differences between exports and imports results.

To test this hypothesis we work with the data at the level of 21 HS sections and ask whether, for each HS section, the decline in trade is larger in markets that account for a smaller share of total exporters' exports. The test is done in two stages. First, we run a triple difference for each HS section and flow (exports, imports) separately estimating a TCA effect for each EU27 member state:

$$Y_{ijthf} = \exp \left[ \alpha_{ithf} + \alpha_{jthf} + \alpha_{ijhf} + \sum_{j=1}^{27} \beta_{jhf} (EU_j \times UK_i \times Post_t \times a_j) \right] + \epsilon_{ijthf} \quad (10)$$

where  $a_j$  is a dummy for each partner,  $h$  stands for the HS section and  $f$  for flow (exports, imports). Estimating equation (10) by PPML for each HS section and flow separately we obtain a TCA effect  $\tilde{\beta}_{jhf}$  for each HS section  $h$ , each EU27 member  $j$  and each flow  $f$ . In total there are 21 HS section, 27 EU members and 2 flows, so we should obtain 1,134 coefficients. However, for some EU27 member-HS section combination where trade is very sparse we cannot estimate a coefficient due to insufficient data or no trade (e.g., UK imports of Arms and ammunitions from Bulgaria). In total we are able to estimate 1,112 partner-section coefficients, hence losing 22 coefficients.

The second stage involves the regression of the  $\tilde{\beta}_{jhf}$  coefficients on the exports share  $w_{jhf}$  and HS section by flow dummies. The exports share  $w_{jh,exports} = \frac{X_{jh}}{X_h}$  are defined as the ratio of UK exports of HS section  $h$  to EU27 member  $j$  over total UK exports of HS section  $h$  to the world. Similarly, for imports  $w_{jh,imports}$  are defined as  $j$ 's exports of HS section  $h$  to the UK over total  $j$ 's exports of HS section  $h$  to the world. The shares are computed with data for total exports over the years 2017-19. The regression that we estimate is:

$$\tilde{\beta}_{jhf} = a + a_{hf} + \delta w_{jhf} + \epsilon_{jhf} \quad (11)$$

If our hypothesis is correct, then we should expect  $\delta > 0$ . It is important to notice that our test is done within HS section, using variation in shares across partners. This means that differences in trade policies that could influence the shares are controlled for, given that the trade policy is the same for all EU27 member states.

The summary statistics for the first stage results  $\tilde{\beta}_{jhf}$  and the exports shares  $w_{jhf}$  to be used in the second stage are reported in Table 9. For UK imports from the EU, the average TCA effects across HS sections and partners is -0.536, while for UK exports to the EU the average effect is -0.244. The average exports share for EU exports to the UK is 0.06 while for UK exports to the EU is 0.02. These are simple averages across HS sections and partners hence they do not take into account the size of each sector/market.<sup>16</sup>

Table 9: TCA effects and exports shares by HS section and partners

|                     | Mean   | sd    | Median | Min    | Max   | N   |
|---------------------|--------|-------|--------|--------|-------|-----|
| $\tilde{b}$ imports | -0.536 | 1.160 | -0.316 | -9.386 | 4.215 | 546 |
| $\tilde{b}$ exports | -0.244 | 1.061 | -0.242 | -3.593 | 8.140 | 567 |
| Share UK imports    | 0.060  | 0.093 | 0.038  | 0.000  | 0.820 | 546 |
| Share UK exports    | 0.020  | 0.042 | 0.004  | 0.000  | 0.477 | 567 |

The second stage results are reported in Table 10. In the first column we report the results for the estimation of model (11) by OLS. To account for the uncertainty in the estimation of  $\tilde{\beta}_{jhs}$  given by the first stage, in column two we estimate model (11) by weighted least squares with weights given by the t-stats of the first stage coefficients. In both cases we find a positive and statistically significant coefficient on the export share, suggesting that the importance of the destination market in total exports is a driver of the TCA effects. In columns 3-4 we also exclude HS section 21 (Arts and antiques) because of the special nature of these products. In this case the regression coefficients are larger and more precisely estimated.

<sup>16</sup>Some of the EU markets such as Malta or Cyprus are small, while the UK can be a large destination for exports of these small EU countries. This explains the fact that the average exports share is slightly larger for EU27 exports to the UK than the reverse. On the other hand, when we consider the share of total exports the EU27 is a very large market for the UK, while the UK is relatively small for the EU27.

Table 10: Regression of TCA effects on partner-section exports shares

|                 | All sections         |                      | Ex. Arts and antiques |                      |
|-----------------|----------------------|----------------------|-----------------------|----------------------|
|                 | (1)                  | (2)                  | (3)                   | (4)                  |
|                 | OLS                  | WLS                  | OLS                   | WLS                  |
| Exports share   | 0.555**<br>(0.279)   | 2.025*<br>(1.145)    | 0.839***<br>(0.233)   | 3.007***<br>(1.023)  |
| Constant        | -0.409***<br>(0.034) | -0.808***<br>(0.097) | -0.470***<br>(0.029)  | -0.933***<br>(0.093) |
| Section-flow FE | Yes                  | Yes                  | Yes                   | Yes                  |
| Observations    | 1,113                | 1,113                | 1,062                 | 1,062                |

Robust standard errors in parenthesis. \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ . The table reports the results of the regressions of the partner-section TCA effects estimated with PPML on the exports share. The regression pool together TCA effects on exports and imports and include section-flow fixed effects. Exports shares for the TCA effects on exports are defined as UK exports of HS section to a EU27 member of total UK exports of HS section over 2017-19. Exports shares for the TCA effects on imports are defined as EU27 member exports of HS section to the UK over total EU27 member exports of HS section to the world over 2017-19.

How do these results relate to the differences in the TCA effects on total exports and imports of Section 4.2? Using the OLS estimate of column one we have the following equation for the TCA effect:

$$\text{TCA effect} = -0.409 + 0.555 \times \text{Exports share} \quad (12)$$

The exports share of the EU27 to the UK over 2017-19 is 0.063. Using this value, the equation predicts a TCA effect of -0.374 with 95% CI of [-0.425, -0.324] and the triple difference estimation of Table (3) gives a coefficient of -0.323 for UK imports from the EU. This is remarkably close. For UK exports to the EU the exports share over 2017-19 is 0.467, hence the model predicts a TCA effect of -0.15. This is larger than the DDD estimate of -0.036 but within the 95% confidence interval of the predictions of model (12) which is [-0.367, 0.066] and includes the zero.<sup>17</sup>

Overall, these results provide support for the hypothesis that the asymmetries in the TCA impacts for UK exports to and imports from the EU can be explained by the relative importance of the destination market for each exporters. However, it is important to note, that this does not constitute a detailed explanation of all the factors driving the TCA effects, which is beyond the scope of this paper. This is because the export shares are determined by a range of factors determining market access barriers ranging from policy to market structure, as well as geographical factors such as distance, and technological factors impacting on the organization of production and supply chains.

<sup>17</sup>Using the regressions of column 3 of Table 10 without HS section 21 we obtain predictions for total imports of -0.417 [-0.459, -0.376] and -0.078 [-0.254, 0.098] for total exports.

## 5 Conclusions

Our analysis of monthly goods trade data shows no evidence of a negative impact on UK trade with the EU in the period following the referendum of 2016. In contrast we find that the introduction of the Trade and Cooperation Agreement in January 2021 had a strong effect on UK-EU trade in 2021. Our results are in line with the dynamic general equilibrium simulation of Steinberg (2019), which show negligible impacts of referendum-induced uncertainty on UK-EU trade but substantial TCA effects, and with UK imports slightly more affected than UK exports. We find significant differences between UK exports to and imports from the EU. UK exports, in aggregate, were strongly affected in January 2021 only to recover to normal levels in the following months. The substantial decline in January 2021 is almost certainly attributable to the last-minute nature of the deal, and the lack of time firms had to prepare for the agreement that was put in place. An interesting feature of the results is the subsequent recovery in exports, which was probably not anticipated by most commentators and no doubt more work will need to be undertaken in due course to understand this better. In contrast, UK imports have been consistently affected in all the months of 2021 and have not shown signs of recovery. While compositional differences between imports and exports could in theory explain the asymmetric changes our results indicate clearly that this is not the case. More research is needed on understanding better the drivers of the changes in trade and the asymmetries, such as for example the role of additional bureaucracy, customs procedures, rules of origin, or regulatory requirements. A central plausible explanation for these differential effects of the impacts of exports and imports is the relative importance of the EU market for UK firms, in comparison to the importance of the UK market for EU firms. This meant that it was more important for UK firms to maintain access to the EU, than it was for EU firms to maintain access to the UK. Using exports share, we provide some evidence in support of this hypothesis. While UK exports seems to have responded to new trade barriers and are heterogeneous across products, the negative effects on imports are very similar across products.

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## A Appendix

### A.1 Difference between PPML and log-linear OLS

In the main body of the text we argued in favour of the PPML estimator rather than the log-linear OLS estimator on the ground that the PPML estimator allows us to estimate the proportional effect of the TCA on total exports to the EU, while the log-linear OLS would give us an average effect across EU countries. While both estimators provide an estimate of the proportional treatment effect, the interpretation of coefficients is different. The difference is due to the first order conditions of the two estimators. For the PPML, the FOC is in levels, hence the effect is estimated on the arithmetic average of the dependent variable. The FOC of the log-linear OLS is on the log of the dependent variable, hence the effect is measured on the geometric average. For the difference-in-differences estimation of UK exports to the EU, the PPML coefficient equals the following quantity:

$$\tilde{b}_{PPML} = \ln \left[ \frac{\bar{Y}_{EU,Post} / \bar{Y}_{EU,Pre}}{\bar{Y}_{Control,Post} / \bar{Y}_{Control,Pre}} \right]$$

where  $\bar{Y}_{EU,Post} = (N_{EU}T_{Post})^{-1} \sum_{i \in EU} \sum_{t \in Post} Y_{it}$  is average UK exports to the EU in 2021 (across 27 member states) and  $\bar{Y}_{EU,Pre} = (N_{EU}T_{Pre})^{-1} \sum_{i \in EU} \sum_{t \in Pre} Y_{it}$  is average UK exports to the EU in the pre-TCA period, with  $N_{EU} = 27$  being the number of countries in the EU and  $T_{Post}$  ( $T_{Pre}$ ) the number of periods in the post (pre) TCA period. At the denominator we have the same quantities for exports to the control group. Since we have a balanced panel data, the number of countries within EU and control group are the same both pre and post TCA, and the number of periods are the same pre/post TCA for the EU and control groups, these quantities cancel out so the PPML coefficient effectively measures the proportional change in total trade:

$$\tilde{b}_{PPML} = \ln \left[ \frac{Y_{EU,Post} / Y_{EU,Pre}}{Y_{Control,Post} / Y_{Control,Pre}} \right]$$

Hence, the difference-in-differences estimated with the PPML estimator allows us to retrieve the TCA effect on total trade with the EU.

On the other hand, the OLS difference-in-differences coefficient is:

$$\tilde{b}_{OLS} = (\overline{\ln Y_{EU,Post}} - \overline{\ln Y_{EU,Pre}}) - (\overline{\ln Y_{Control,Post}} - \overline{\ln Y_{Control,Pre}})$$

where  $\overline{\ln Y_{EU,Post}}$  is the geometric average of UK exports to the EU in the post-TCA period:

$$\overline{\ln Y_{EU,Post}} = \frac{1}{N_{EU}T_{Post}} \sum_{i \in EU} \sum_{t \in Post} \ln Y_{it} = \ln \left[ \left( \prod_{i \in EU} \prod_{t \in Post} Y_{it} \right)^{1/N_{EU}T_{Post}} \right] = \ln \left( \bar{Y}^G_{EU,Post} \right)$$

where  $\bar{Y}^G_{EU,Post}$  is the geometric average. Hence the log-linear OLS difference-in-differences coefficient is:

$$\tilde{b}_{OLS} = \ln \left[ \frac{\overline{Y^G}_{EU,Post} / \overline{Y^G}_{EU,Pre}}{\overline{Y^G}_{Control,Post} / \overline{Y^G}_{Control,Pre}} \right]$$

Given the multiplicative nature of the geometric average, a proportional change in one of the elements of the average of the average will have the same weight. This means that a change in UK exports to Malta or to Germany have the same importance. Differently, the PPML estimator gives more importance to larger flows.

## A.2 Estimation of DD and DDD with product-level data

Given that Covid-19 affected some products more than others, one might be tempted to estimate aggregate TCA effects using product-level data so to control for product-time specific shocks. For the difference-in-difference estimation, this would make no difference. This is a property of the PPML estimator. Using the PPML estimator for a difference-in-differences on a pooled product dataset to estimate the treatment effect (with binary treatment) would give exactly the same answer if the estimation was carried out on total trade data aggregating products – after conditioning on the proper set of fixed effects. Consider a dataset of UK exports with the following dimension: partners ( $j$ ), products ( $h$ ) and time ( $t$ ). Treatment affects EU countries, and treatment can be heterogeneous across products. Then consider the following regression estimated by PPML:

$$Y_{jht} = \exp [a_{jh} + a_{ht} + b(EU_j \times D_t)] + e_{jht} \quad (13)$$

Where  $a_{jh}$  are partner-product FE (identifying cross-sectional units) and  $a_{ht}$  are product-time FE – which account for product-specific time shocks. Then  $EU_j=1$  for EU countries and zero for non-EU, and  $D_t$  is a dummy variable for the post-treatment period. Using the PPML estimator, the coefficient  $b$  gives the proportional response on the arithmetic average of  $Y_{jht}$ . This means that larger products will receive larger weights and so in the presence of heterogeneous treatment effects across products, the estimate of  $b$  will be driven by treatment effects of large products. Alternatively, we can collapse data at the country-product level and estimate the following:

$$Y_{jt} = \exp [a_j + a_t + b(EU_j \times D_t)] + e_{jt} \quad (14)$$

Which is our DD equation (1). With a panel product-level dataset, equations (13) and (14) would yield exactly the same treatment effect coefficient  $b$ .

Differently, if the DD regression was estimated by OLS after taking the log of  $Y$ , using data by product or total trade would give different treatment effects *in the presence of treatment heterogeneity across products*. This is because, as explained in the Section A.1 of the Appendix, the log-linear OLS gives the proportional response on the geometric average.

The same would be true for a triple difference regression if exporter-time and importer-time FE were not included – which at the product level becomes exporter-product-time and importer-product-time. The use of these exporter-time and importer-time FE breaks the equality of coefficients between PPML regressions at the product or total level. However, the results would remain very similar, and that is again because of the weighting scheme of the PPML estimator.

Hence, we believe that the estimation using product-level data will have any material effect on our treatment effects – exactly no differences for DD and only minor ones for DDD because of exporter-time and importer-time FE.

To prove our point, we estimate the following DDD regression with data at the HS 2-digit level:

$$Y_{ijht} = \exp[\alpha_{iht} + \alpha_{jht} + \alpha_{ijh} + \beta(EU_j \times UK_i \times Post_t)] + \epsilon_{ijht}$$

where  $\alpha_{iht}$ ,  $\alpha_{jht}$  and  $\alpha_{ijh}$  are reporter-HS2-time, partner-HS2-time and reporter-partner-HS2 fixed effects. Results are reported in Table 11. As we can see, results are virtually identical to the baseline DDD results of Table (3).

Table 11: DDD PPML pooled HS2, 2017-21

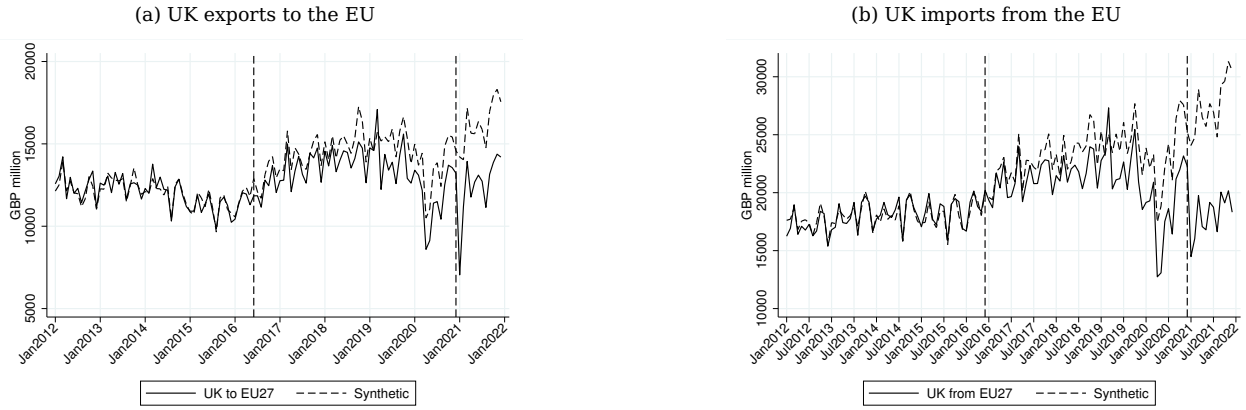
|                                | (1)<br>Exports    | (2)<br>Imports       |
|--------------------------------|-------------------|----------------------|
| $UK_i \times EU_j \times 2021$ | -0.039<br>(0.036) | -0.294***<br>(0.034) |
| Reporter-HS2-time FE           | Yes               | Yes                  |
| Partner-HS2-time FE            | Yes               | Yes                  |
| Reporter-partner-HS2 FE        | Yes               | Yes                  |
| Observations                   | 4,821,600         | 4,607,159            |

Robust standard errors clustered at the reporter-partner-HS2 level in parenthesis. \* p < 0.10, \*\* p < 0.05, \*\*\* p < 0.01.

### A.3 Additional results for 2016 referendum

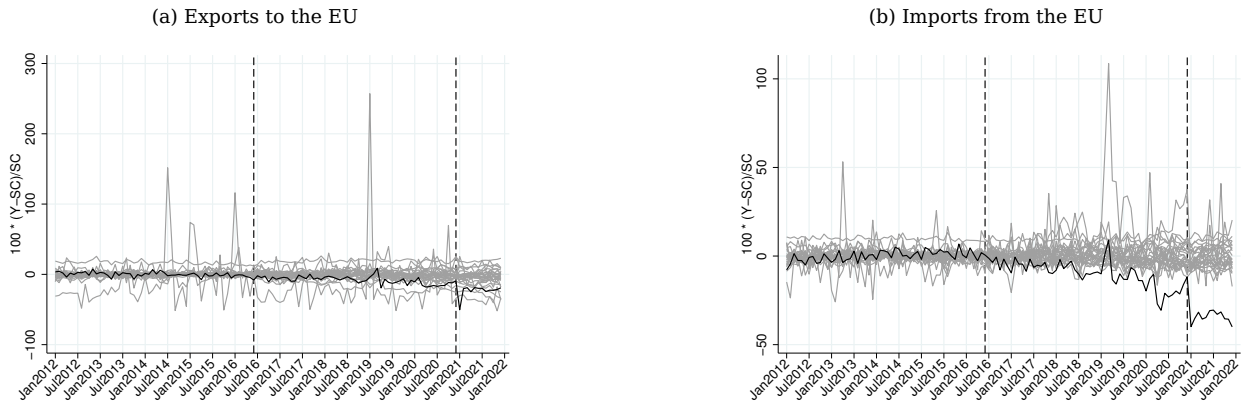
Synthetic control matched over the period Jan2012-Jun2016. Excludes Croatia as it joined the EU in 2013.

Figure 11: UK-EU trade SCM result



The figure reports the SC estimates for UK exports to and imports from the EU27. The pre-treatment period is Jan2012-Jun2016, matching is performed over all pre-treatment values of the outcome variable.

Figure 12: UK-EU trade SCM result



The figure reports placebo of the SC estimates for UK trade with the EU27. The pre-treatment period is Jan2012-Jun2016, matching is performed over all pre-treatment values of the outcome variable. The vertical axis measures percentage deviations of the SC from the outcome variable Y. The black solid line represents UK-EU27 trade while the grey lines are trade of each EU27 member with the remaining EU countries.

#### A.4 Robustness to DD and DDD specification for TCA effects

Table 12: DD results total trade ex. gold controlling for Covid-19

|                     | Exports              | Exports              | Imports              | Imports              | Exports             | Exports              | Imports              | Imports              |
|---------------------|----------------------|----------------------|----------------------|----------------------|---------------------|----------------------|----------------------|----------------------|
| EU×2021             | -0.017<br>(0.053)    |                      | -0.364***<br>(0.072) |                      | -0.016<br>(0.056)   |                      | -0.367***<br>(0.071) |                      |
| Log Covid-19 deaths | -0.015***<br>(0.005) | -0.014**<br>(0.005)  | -0.021*<br>(0.012)   | -0.025**<br>(0.011)  |                     |                      |                      |                      |
| EU×Jan×2021         |                      | -0.528***<br>(0.065) |                      | -0.246***<br>(0.056) |                     | -0.531***<br>(0.070) |                      | -0.249***<br>(0.054) |
| EU×Feb×2021         |                      | -0.069<br>(0.061)    |                      | -0.330***<br>(0.088) |                     | -0.074<br>(0.063)    |                      | -0.333***<br>(0.081) |
| EU×Mar×2021         |                      | -0.085*<br>(0.046)   |                      | -0.298**<br>(0.129)  |                     | -0.086*<br>(0.049)   |                      | -0.300**<br>(0.130)  |
| EU×Apr×2021         |                      | 0.006<br>(0.053)     |                      | -0.320***<br>(0.093) |                     | 0.006<br>(0.052)     |                      | -0.317***<br>(0.087) |
| EU×May×2021         |                      | -0.001<br>(0.051)    |                      | -0.316***<br>(0.065) |                     | 0.002<br>(0.050)     |                      | -0.323***<br>(0.055) |
| EU×Jun×2021         |                      | 0.083<br>(0.059)     |                      | -0.292***<br>(0.048) |                     | 0.083<br>(0.057)     |                      | -0.305***<br>(0.046) |
| EU×Jul×2021         |                      | 0.037<br>(0.065)     |                      | -0.295***<br>(0.082) |                     | 0.043<br>(0.065)     |                      | -0.300***<br>(0.077) |
| EU×Aug×2021         |                      | 0.012<br>(0.072)     |                      | -0.261***<br>(0.069) |                     | 0.014<br>(0.073)     |                      | -0.277***<br>(0.066) |
| EU×Sep×2021         |                      | 0.045<br>(0.083)     |                      | -0.372***<br>(0.084) |                     | 0.046<br>(0.083)     |                      | -0.385***<br>(0.079) |
| EU×Oct×2021         |                      | 0.009<br>(0.107)     |                      | -0.465***<br>(0.143) |                     | 0.008<br>(0.107)     |                      | -0.464***<br>(0.145) |
| EU×Nov×2021         |                      | 0.088<br>(0.094)     |                      | -0.454***<br>(0.144) |                     | 0.091<br>(0.094)     |                      | -0.448***<br>(0.145) |
| EU×Dec×2021         |                      | 0.077<br>(0.078)     |                      | -0.606***<br>(0.135) |                     | 0.079<br>(0.077)     |                      | -0.596***<br>(0.138) |
| Log Covid-19 cases  |                      |                      |                      |                      | -0.012**<br>(0.005) | -0.011**<br>(0.006)  | -0.024**<br>(0.011)  | -0.029***<br>(0.011) |
| Partner FE          | Yes                  | No                   | Yes                  | No                   | Yes                 | No                   | Yes                  | No                   |
| Time FE             | Yes                  | Yes                  | Yes                  | Yes                  | Yes                 | Yes                  | Yes                  | Yes                  |
| Partner-month FE    | No                   | Yes                  | No                   | Yes                  | No                  | Yes                  | No                   | Yes                  |
| Observations        | 2640                 | 2640                 | 2640                 | 2640                 | 2640                | 2640                 | 2640                 | 2640                 |

Robust standard errors clustered at the exporter-importer level in parenthesis. \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001.

Table 13: DD results total trade ex. gold controlling for freight cost

|                     | Exports              | Exports              | Imports              | Imports              | Exports              | Exports              | Imports              | Imports              |
|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| EU×Year=2021        | -0.215***<br>(0.068) |                      | -0.299***<br>(0.095) |                      | -0.204***<br>(0.067) |                      | -0.273***<br>(0.095) |                      |
| ln(dist)×ln(FBX)    | -0.059***<br>(0.020) | -0.058***<br>(0.021) | 0.016<br>(0.040)     | 0.003<br>(0.044)     | -0.054***<br>(0.018) | -0.055***<br>(0.020) | 0.016<br>(0.028)     | -0.001<br>(0.029)    |
| EU×Year=2021×Jan    |                      | -0.658***<br>(0.072) |                      | -0.298***<br>(0.061) |                      | -0.648***<br>(0.064) |                      | -0.245***<br>(0.047) |
| EU×Year=2021×Feb    |                      | -0.211***<br>(0.076) |                      | -0.384***<br>(0.081) |                      | -0.198***<br>(0.072) |                      | -0.328***<br>(0.057) |
| EU×Year=2021×Mar    |                      | -0.233***<br>(0.074) |                      | -0.332***<br>(0.103) |                      | -0.222***<br>(0.073) |                      | -0.298***<br>(0.092) |
| EU×Year=2021×Apr    |                      | -0.154**<br>(0.064)  |                      | -0.350***<br>(0.091) |                      | -0.142**<br>(0.065)  |                      | -0.320***<br>(0.090) |
| EU×Year=2021×May    |                      | -0.181**<br>(0.080)  |                      | -0.348***<br>(0.105) |                      | -0.172**<br>(0.082)  |                      | -0.317***<br>(0.105) |
| EU×Year=2021×Jun    |                      | -0.096<br>(0.081)    |                      | -0.304***<br>(0.088) |                      | -0.091<br>(0.082)    |                      | -0.294***<br>(0.072) |
| EU×Year=2021×Jul    |                      | -0.173**<br>(0.086)  |                      | -0.306*<br>(0.169)   |                      | -0.171**<br>(0.084)  |                      | -0.298**<br>(0.147)  |
| EU×Year=2021×Aug    |                      | -0.208**<br>(0.092)  |                      | -0.277**<br>(0.139)  |                      | -0.206**<br>(0.092)  |                      | -0.264**<br>(0.124)  |
| EU×Year=2021×Sep    |                      | -0.165<br>(0.112)    |                      | -0.377**<br>(0.188)  |                      | -0.167<br>(0.113)    |                      | -0.375**<br>(0.162)  |
| EU×Year=2021×Oct    |                      | -0.223*<br>(0.116)   |                      | -0.436<br>(0.271)    |                      | -0.211*<br>(0.121)   |                      | -0.470**<br>(0.210)  |
| Log Covid-19 deaths |                      |                      |                      |                      | -0.015***<br>(0.005) | -0.013***<br>(0.005) | -0.025**<br>(0.010)  | -0.026**<br>(0.010)  |
| Partner FE          | Yes                  | No                   | Yes                  | No                   | Yes                  | No                   | Yes                  | No                   |
| Time FE             | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Partner-month FE    | No                   | Yes                  | No                   | Yes                  | No                   | Yes                  | No                   | Yes                  |
| Observations        | 2552                 | 2552                 | 2552                 | 2552                 | 2552                 | 2552                 | 2552                 | 2552                 |

Robust standard errors clustered at the exporter-importer level in parenthesis. \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001.

Table 14: DD results total trade ex. gold controlling for Covid-19 and distance-by-time

|                      | Exports             | Exports              | Imports              | Imports              | Exports              | Exports              | Imports              | Imports              |
|----------------------|---------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| EU × Year=2021       | -0.254**<br>(0.103) |                      | -0.496***<br>(0.104) |                      | -0.251**<br>(0.105)  |                      | -0.475***<br>(0.102) |                      |
| EU × Year=2021 × Jan |                     | -0.736***<br>(0.094) |                      | -0.168***<br>(0.058) |                      | -0.734***<br>(0.096) |                      | -0.110*<br>(0.060)   |
| EU × Year=2021 × Feb |                     | -0.300*<br>(0.154)   |                      | -0.127<br>(0.088)    |                      | -0.290*<br>(0.153)   |                      | -0.067<br>(0.094)    |
| EU × Year=2021 × Mar |                     | -0.202*<br>(0.118)   |                      | -0.079<br>(0.153)    |                      | -0.192<br>(0.122)    |                      | -0.026<br>(0.159)    |
| EU × Year=2021 × Apr |                     | -0.206<br>(0.145)    |                      | -0.201**<br>(0.096)  |                      | -0.195<br>(0.146)    |                      | -0.159<br>(0.099)    |
| EU × Year=2021 × May |                     | -0.188*<br>(0.113)   |                      | -0.360***<br>(0.088) |                      | -0.174<br>(0.120)    |                      | -0.310***<br>(0.100) |
| EU × Year=2021 × Jun |                     | -0.156<br>(0.138)    |                      | -0.298***<br>(0.056) |                      | -0.147<br>(0.140)    |                      | -0.266***<br>(0.065) |
| EU × Year=2021 × Jul |                     | -0.210***<br>(0.078) |                      | -0.563***<br>(0.077) |                      | -0.208**<br>(0.083)  |                      | -0.538***<br>(0.065) |
| EU × Year=2021 × Aug |                     | -0.242**<br>(0.103)  |                      | -0.308***<br>(0.118) |                      | -0.246**<br>(0.106)  |                      | -0.297***<br>(0.104) |
| EU × Year=2021 × Sep |                     | -0.043<br>(0.156)    |                      | -0.617***<br>(0.187) |                      | -0.047<br>(0.160)    |                      | -0.605***<br>(0.172) |
| EU × Year=2021 × Oct |                     | -0.359**<br>(0.144)  |                      | -0.953***<br>(0.228) |                      | -0.362**<br>(0.144)  |                      | -0.969***<br>(0.235) |
| EU × Year=2021 × Nov |                     | -0.272***<br>(0.101) |                      | -0.782***<br>(0.241) |                      | -0.276***<br>(0.100) |                      | -0.798***<br>(0.246) |
| EU × Year=2021 × Dec |                     | -0.213<br>(0.168)    |                      | -1.030***<br>(0.201) |                      | -0.216<br>(0.169)    |                      | -1.048***<br>(0.206) |
| Log Covid-19 deaths  |                     |                      |                      |                      | -0.015***<br>(0.005) | -0.014***<br>(0.005) | -0.022*<br>(0.012)   | -0.027***<br>(0.009) |
| Partner FE           | Yes                 | No                   | Yes                  | No                   | Yes                  | No                   | Yes                  | No                   |
| Time FE              | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Partner-month FE     | No                  | Yes                  | No                   | Yes                  | No                   | Yes                  | No                   | Yes                  |
| ln(dist) × time FE   | Yes                 | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Observations         | 2640                | 2640                 | 2640                 | 2640                 | 2640                 | 2640                 | 2640                 | 2640                 |

Robust standard errors clustered at the exporter-importer level in parenthesis. \* p < 0.05, \*\* p < 0.01, \*\*\* p < 0.001.



Table 15: DDD results total trade ex. gold controlling for freight cost by distance

|                           | Exports           | Exports              | Imports              | Imports              | Exports           | Exports              | Imports              | Imports              |
|---------------------------|-------------------|----------------------|----------------------|----------------------|-------------------|----------------------|----------------------|----------------------|
| UK×EU×2021                | -0.048<br>(0.048) |                      | -0.314***<br>(0.044) |                      | -0.036<br>(0.049) |                      | -0.323***<br>(0.044) |                      |
| ln(dist)×ln(FBX)          | -0.003<br>(0.007) | -0.001<br>(0.007)    | 0.000<br>(0.006)     | 0.004<br>(0.006)     |                   |                      |                      |                      |
| UK×EU×Jan×2021            |                   | -0.520***<br>(0.065) |                      | -0.381***<br>(0.060) |                   | -0.520***<br>(0.065) |                      | -0.382***<br>(0.061) |
| UK×EU×Feb×2021            |                   | -0.033<br>(0.064)    |                      | -0.412***<br>(0.080) |                   | -0.033<br>(0.064)    |                      | -0.412***<br>(0.080) |
| UK×EU×Mar×2021            |                   | -0.118***<br>(0.045) |                      | -0.331***<br>(0.093) |                   | -0.118***<br>(0.046) |                      | -0.331***<br>(0.093) |
| UK×EU×Apr×2021            |                   | 0.009<br>(0.054)     |                      | -0.374***<br>(0.065) |                   | 0.009<br>(0.055)     |                      | -0.375***<br>(0.064) |
| UK×EU×May×2021            |                   | -0.013<br>(0.045)    |                      | -0.359***<br>(0.050) |                   | -0.013<br>(0.046)    |                      | -0.360***<br>(0.049) |
| UK×EU×Jun×2021            |                   | 0.055<br>(0.049)     |                      | -0.292***<br>(0.055) |                   | 0.055<br>(0.049)     |                      | -0.292***<br>(0.055) |
| UK×EU×Jul×2021            |                   | 0.024<br>(0.054)     |                      | -0.314***<br>(0.060) |                   | 0.024<br>(0.054)     |                      | -0.313***<br>(0.060) |
| UK×EU×Aug×2021            |                   | 0.017<br>(0.070)     |                      | -0.201***<br>(0.062) |                   | 0.017<br>(0.070)     |                      | -0.200***<br>(0.063) |
| UK×EU×Sep×2021            |                   | 0.016<br>(0.072)     |                      | -0.278***<br>(0.050) |                   | 0.016<br>(0.072)     |                      | -0.277***<br>(0.050) |
| UK×EU×Oct×2021            |                   | -0.036<br>(0.095)    |                      | -0.317***<br>(0.082) |                   | -0.037<br>(0.096)    |                      | -0.316***<br>(0.083) |
| UK×EU×Nov×2021            |                   |                      |                      |                      |                   | 0.047<br>(0.085)     |                      | -0.268***<br>(0.060) |
| UK×EU×Dec×2021            |                   |                      |                      |                      |                   | 0.002<br>(0.066)     |                      | -0.369***<br>(0.075) |
| Partner-time FE           | Yes               | Yes                  | Yes                  | Yes                  | Yes               | Yes                  | Yes                  | Yes                  |
| Reporter-time FE          | Yes               | Yes                  | Yes                  | Yes                  | Yes               | Yes                  | Yes                  | Yes                  |
| Pair FE                   | Yes               | No                   | Yes                  | No                   | Yes               | No                   | Yes                  | No                   |
| Pair-month FE             | No                | Yes                  | No                   | Yes                  | No                | Yes                  | No                   | Yes                  |
| ln( <i>dist</i> )×time-FE | No                | No                   | No                   | No                   | Yes               | Yes                  | Yes                  | Yes                  |
| Observations              | 69890             | 69890                | 69890                | 69890                | 72300             | 72300                | 72300                | 72300                |

Robust standard errors clustered at the exporter-importer level in parenthesis. \*

p &lt; 0.05, \*\* p &lt; 0.01, \*\*\* p &lt; 0.001.

To see whether the inclusion of FTA partners in the control group is driving results, both for the referendum and the TCA effects, we exclude the FTA partners from the control group (these are: Canada, Switzerland, Chile, Colombia, Israel, Iceland, Japan, South Korea, Mexico, Norway, Turkey and South Africa). Results are reported in Table 16. Coefficient estimates and significance level is very similar to the baseline results reported in the main text.

Table 16: DDD results excluding FTA partner

|                  | Exports           | Exports              | Imports              | Imports              |
|------------------|-------------------|----------------------|----------------------|----------------------|
| UK×EU×2021       | -0.041<br>(0.051) |                      | -0.274***<br>(0.050) |                      |
| UK×EU×Jan×2021   |                   | -0.524***<br>(0.071) |                      | -0.408***<br>(0.071) |
| UK×EU×Feb×2021   |                   | -0.020<br>(0.061)    |                      | -0.463***<br>(0.094) |
| UK×EU×Mar×2021   |                   | -0.143***<br>(0.044) |                      | -0.338***<br>(0.115) |
| UK×EU×Apr×2021   |                   | -0.001<br>(0.054)    |                      | -0.325***<br>(0.072) |
| UK×EU×May×2021   |                   | -0.029<br>(0.044)    |                      | -0.309***<br>(0.048) |
| UK×EU×Jun×2021   |                   | 0.074<br>(0.052)     |                      | -0.307***<br>(0.063) |
| UK×EU×Jul×2021   |                   | -0.012<br>(0.047)    |                      | -0.215***<br>(0.063) |
| UK×EU×Aug×2021   |                   | -0.015<br>(0.070)    |                      | -0.142*<br>(0.077)   |
| UK×EU×Sep×2021   |                   | -0.004<br>(0.077)    |                      | -0.196***<br>(0.038) |
| UK×EU×Oct×2021   |                   | -0.031<br>(0.105)    |                      | -0.153***<br>(0.045) |
| UK×EU×Nov×2021   |                   | 0.103<br>(0.074)     |                      | -0.187***<br>(0.062) |
| UK×EU×Dec×2021   |                   | -0.007<br>(0.071)    |                      | -0.313***<br>(0.064) |
| Partner-time FE  | Yes               | Yes                  | Yes                  | Yes                  |
| Reporter-time FE | Yes               | Yes                  | Yes                  | Yes                  |
| Pair FE          | Yes               | No                   | Yes                  | No                   |
| Pair-month FE    | No                | Yes                  | No                   | Yes                  |
| Observations     | 55500             | 55500                | 55500                | 55500                |

Robust standard errors clustered at the exporter-importer level in parenthesis. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The table reports the DDD estimates for UK exports to and imports from the EU excluding FTA partners from the control group.

Table 17: DDD results excluding Covid-related products

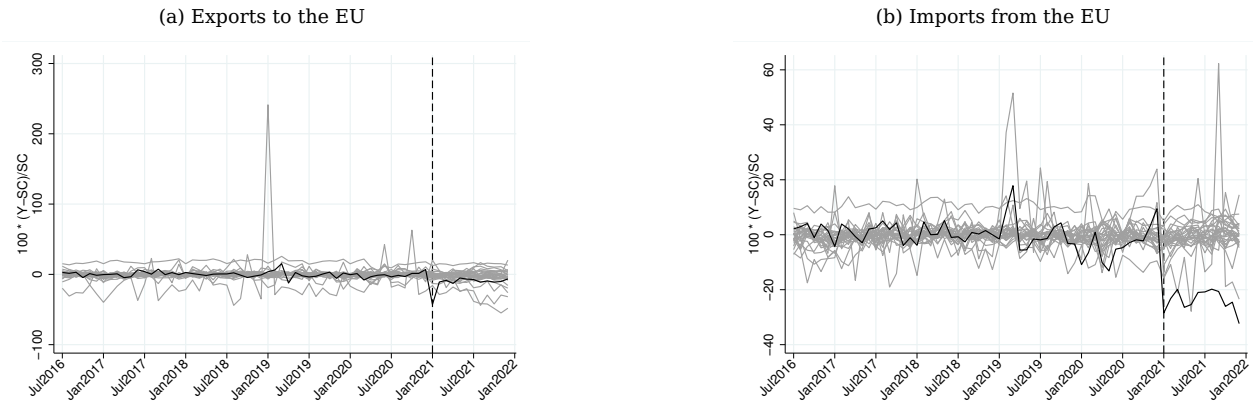
|                  | Exports           | Exports              | Imports              | Imports              |
|------------------|-------------------|----------------------|----------------------|----------------------|
| UK×EU×2021       | -0.050<br>(0.043) |                      | -0.295***<br>(0.045) |                      |
| UK×EU×Jan×2021   |                   | -0.558***<br>(0.059) |                      | -0.347***<br>(0.052) |
| UK×EU×Feb×2021   |                   | -0.066<br>(0.059)    |                      | -0.354***<br>(0.069) |
| UK×EU×Mar×2021   |                   | -0.136***<br>(0.041) |                      | -0.277***<br>(0.083) |
| UK×EU×Apr×2021   |                   | 0.001<br>(0.057)     |                      | -0.340***<br>(0.068) |
| UK×EU×May×2021   |                   | -0.022<br>(0.048)    |                      | -0.315***<br>(0.057) |
| UK×EU×Jun×2021   |                   | 0.061<br>(0.051)     |                      | -0.256***<br>(0.064) |
| UK×EU×Jul×2021   |                   | 0.029<br>(0.054)     |                      | -0.293***<br>(0.067) |
| UK×EU×Aug×2021   |                   | -0.023<br>(0.065)    |                      | -0.171***<br>(0.061) |
| UK×EU×Sep×2021   |                   | 0.013<br>(0.072)     |                      | -0.270***<br>(0.051) |
| UK×EU×Oct×2021   |                   | -0.081<br>(0.093)    |                      | -0.313***<br>(0.075) |
| UK×EU×Nov×2021   |                   | 0.051<br>(0.077)     |                      | -0.246***<br>(0.060) |
| UK×EU×Dec×2021   |                   | 0.001<br>(0.062)     |                      | -0.370***<br>(0.075) |
| Partner-time FE  | Yes               | Yes                  | Yes                  | Yes                  |
| Reporter-time FE | Yes               | Yes                  | Yes                  | Yes                  |
| Pair FE          | Yes               | No                   | Yes                  | No                   |
| Pair-month FE    | No                | Yes                  | No                   | Yes                  |
| Observations     | 72300             | 72300                | 72300                | 72300                |

Robust standard errors clustered at the exporter-importer level in parenthesis. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The table reports the DDD estimates for UK exports to and imports from the EU excluding Covid19-related products.

## A.5 Synthetic control placebos

Figure 13 reports the placebos for the percentage difference of the actual flow relative to the SC estimate for the UK (black line) and the placebos (in light grey).

Figure 13: UK-EU trade SCM placebo



The figure reports placebo of the SC estimates for UK trade with the EU27. The pre-treatment period is Jul2016-Dec2020, matching is performed over all pre-treatment values of the outcome variable. The vertical axis measures percentage deviations of the SC from the outcome variable Y. The black solid line represents UK-EU27 trade while the grey lines are trade of each EU27 member with the remaining EU countries.

## A.6 Disaggregated SC results by partner and months of 2021

Table 18: Disaggregated SC results for UK exports

| Importer | Jan_Dec          | Jan              | Feb              | Mar              | Apr              | May              | Jun              | Jul              | Aug              | Sep              | Oct              | Nov              | Dec              |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| AT       | -20.98<br>(0.07) | -43.58<br>(0.00) | -20.89<br>(0.11) | -4.38<br>(0.78)  | -9.21<br>(0.56)  | -13.55<br>(0.44) | 3.02<br>(0.81)   | -24.06<br>(0.11) | -27.20<br>(0.11) | -34.08<br>(0.04) | -34.39<br>(0.00) | -5.66<br>(0.63)  | -37.76<br>(0.00) |
| BE       | 0.93<br>(0.63)   | -37.07<br>(0.00) | 21.07<br>(0.15)  | 19.54<br>(0.15)  | 0.23<br>(0.96)   | 9.54<br>(0.67)   | -1.47<br>(0.89)  | 16.93<br>(0.37)  | 2.17<br>(0.89)   | -3.27<br>(0.78)  | -1.84<br>(0.85)  | -6.17<br>(0.70)  | -8.53<br>(0.63)  |
| BG       | -33.88<br>(0.41) | -54.65<br>(0.11) | -45.25<br>(0.11) | -14.56<br>(0.59) | -36.95<br>(0.26) | -44.63<br>(0.22) | -49.71<br>(0.15) | -36.01<br>(0.41) | -29.25<br>(0.48) | -9.89<br>(0.85)  | -53.17<br>(0.07) | -15.36<br>(0.70) | -17.16<br>(0.74) |
| CY       | -37.07<br>(0.22) | -50.00<br>(0.11) | -39.84<br>(0.26) | -28.06<br>(0.19) | -39.09<br>(0.15) | -24.87<br>(0.26) | -35.58<br>(0.22) | -34.52<br>(0.04) | -39.89<br>(0.19) | -51.01<br>(0.07) | -49.08<br>(0.11) | -34.80<br>(0.22) | -18.11<br>(0.37) |
| CZ       | -24.46<br>(0.00) | -56.66<br>(0.00) | -24.00<br>(0.04) | -11.76<br>(0.11) | -24.00<br>(0.04) | -12.89<br>(0.30) | -16.30<br>(0.22) | -21.71<br>(0.15) | -18.79<br>(0.15) | -32.29<br>(0.04) | -27.70<br>(0.07) | -23.35<br>(0.07) | -24.03<br>(0.11) |
| DE       | -18.73<br>(0.00) | -43.05<br>(0.00) | -17.83<br>(0.00) | -10.30<br>(0.04) | -13.56<br>(0.11) | -5.73<br>(0.63)  | -17.88<br>(0.11) | -15.67<br>(0.19) | -23.83<br>(0.07) | -22.19<br>(0.11) | -22.24<br>(0.04) | -16.05<br>(0.22) | -16.43<br>(0.26) |
| DK       | -12.07<br>(0.15) | -39.08<br>(0.00) | -20.18<br>(0.07) | -11.33<br>(0.26) | 0.34<br>(0.89)   | 1.16<br>(0.81)   | -7.19<br>(0.56)  | -1.41<br>(0.81)  | 0.29<br>(0.96)   | -21.87<br>(0.15) | -20.43<br>(0.15) | -4.78<br>(0.67)  | -20.31<br>(0.22) |
| EE       | -8.66<br>(0.44)  | -40.10<br>(0.04) | -3.90<br>(0.70)  | -18.07<br>(0.11) | 13.44<br>(0.44)  | -19.43<br>(0.26) | -25.39<br>(0.07) | -5.85<br>(0.67)  | 22.06<br>(0.11)  | -11.99<br>(0.26) | -18.14<br>(0.15) | -0.41<br>(0.96)  | 3.93<br>(0.85)   |
| ES       | -24.77<br>(0.07) | -39.97<br>(0.00) | -29.97<br>(0.00) | -28.43<br>(0.00) | -25.90<br>(0.00) | -24.07<br>(0.04) | -25.24<br>(0.11) | -16.35<br>(0.37) | -16.12<br>(0.37) | -4.76<br>(0.70)  | -35.97<br>(0.00) | -31.58<br>(0.00) | -18.86<br>(0.37) |
| FI       | -30.98<br>(0.04) | -42.30<br>(0.00) | -25.10<br>(0.11) | -36.73<br>(0.00) | -27.26<br>(0.04) | -32.11<br>(0.04) | -33.10<br>(0.04) | -30.99<br>(0.04) | -32.56<br>(0.04) | -33.30<br>(0.04) | -24.27<br>(0.30) | -38.59<br>(0.00) | -15.48<br>(0.44) |
| FR       | -16.32<br>(0.11) | -50.56<br>(0.00) | -17.19<br>(0.07) | -11.17<br>(0.22) | -22.38<br>(0.04) | -6.36<br>(0.56)  | -6.81<br>(0.52)  | -8.04<br>(0.59)  | -14.02<br>(0.37) | -11.87<br>(0.26) | -15.51<br>(0.26) | -15.47<br>(0.19) | -16.46<br>(0.33) |
| GR       | -34.82<br>(0.33) | -52.37<br>(0.11) | -41.84<br>(0.07) | -33.06<br>(0.15) | -35.91<br>(0.11) | -35.04<br>(0.26) | -25.43<br>(0.33) | -37.42<br>(0.15) | -34.63<br>(0.19) | -23.07<br>(0.30) | -29.25<br>(0.19) | -34.72<br>(0.26) | -35.12<br>(0.30) |

The table reports the SC treatment effects and p-values (in parenthesis) of UK exports for each EU country and month of 2021.

Table 19: Disaggregated SC results for UK exports (continued)

| Importer | Jan_Dec | Jan    | Feb    | Mar    | Apr    | May    | Jun    | Jul    | Aug    | Sep    | Oct    | Nov    | Dec    |
|----------|---------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| HR       | -57.29  | -65.37 | -53.56 | -52.06 | -56.23 | -49.35 | -50.97 | -44.16 | -45.73 | -59.33 | -56.32 | -73.14 | -81.28 |
|          | (0.04)  | (0.07) | (0.15) | (0.00) | (0.15) | (0.15) | (0.00) | (0.22) | (0.07) | (0.15) | (0.07) | (0.04) | (0.00) |
| HU       | -10.22  | -47.42 | -7.91  | 0.09   | -35.11 | -18.19 | -27.65 | 8.46   | 11.81  | 49.42  | -29.93 | -15.05 | -11.21 |
|          | (0.30)  | (0.00) | (0.37) | (0.96) | (0.11) | (0.22) | (0.19) | (0.67) | (0.56) | (0.00) | (0.15) | (0.59) | (0.56) |
| IE       | 7.82    | -41.01 | -12.34 | -4.75  | -1.07  | -2.86  | 30.05  | 2.62   | 8.04   | 20.00  | 21.18  | 22.04  | 51.94  |
|          | (0.48)  | (0.04) | (0.52) | (0.81) | (0.93) | (0.81) | (0.22) | (0.89) | (0.56) | (0.56) | (0.26) | (0.52) | (0.19) |
| IT       | -16.13  | -57.67 | -7.31  | -10.87 | -14.98 | -1.17  | -1.37  | -15.43 | -17.17 | 1.22   | -21.83 | -17.15 | -29.77 |
|          | (0.04)  | (0.00) | (0.30) | (0.22) | (0.11) | (0.81) | (0.93) | (0.15) | (0.15) | (0.81) | (0.11) | (0.19) | (0.04) |
| LT       | 0.15    | -47.24 | 0.20   | 6.82   | 15.84  | -6.14  | 12.73  | 29.44  | 29.89  | -18.02 | -15.47 | 9.55   | -15.84 |
|          | (0.85)  | (0.30) | (0.96) | (0.78) | (0.63) | (0.89) | (0.74) | (0.48) | (0.44) | (0.70) | (0.59) | (0.78) | (0.63) |
| LU       | 97.73   | 74.59  | -11.91 | 82.90  | 16.88  | 57.98  | 258.78 | 53.10  | 65.59  | 143.50 | 48.43  | 110.52 | 272.34 |
|          | (0.07)  | (0.15) | (0.59) | (0.04) | (0.52) | (0.19) | (0.00) | (0.11) | (0.11) | (0.07) | (0.15) | (0.04) | (0.04) |
| LV       | -10.51  | -62.32 | -22.36 | -7.43  | -10.53 | 0.96   | -23.31 | -9.44  | -2.89  | 7.08   | -8.51  | 37.00  | -24.43 |
|          | (0.33)  | (0.00) | (0.15) | (0.81) | (0.70) | (0.96) | (0.41) | (0.67) | (0.81) | (0.74) | (0.63) | (0.07) | (0.26) |
| MT       | -46.76  | -20.87 | -35.21 | -39.30 | -32.76 | -37.03 | -56.43 | -42.17 | -58.17 | -62.08 | -59.49 | -57.39 | -60.25 |
|          | (0.93)  | (0.70) | (0.74) | (0.52) | (0.63) | (0.74) | (0.59) | (0.67) | (0.52) | (0.52) | (0.41) | (0.59) | (0.48) |
| NL       | -1.63   | -27.69 | 6.37   | -4.86  | -0.86  | 7.21   | -1.60  | 4.15   | -3.38  | -3.15  | 6.76   | -1.29  | -1.24  |
|          | (0.85)  | (0.04) | (0.44) | (0.44) | (0.89) | (0.56) | (0.85) | (0.78) | (0.74) | (0.81) | (0.70) | (0.93) | (0.85) |
| PL       | -32.89  | -67.69 | -28.36 | -22.05 | -35.47 | -24.92 | -31.24 | -29.23 | -19.81 | -36.45 | -32.86 | -37.93 | -28.73 |
|          | (0.00)  | (0.00) | (0.04) | (0.11) | (0.00) | (0.04) | (0.00) | (0.07) | (0.15) | (0.11) | (0.04) | (0.04) | (0.19) |
| PT       | -35.01  | -51.32 | -4.21  | -33.89 | -45.30 | -48.61 | -45.51 | -14.12 | -37.94 | -27.27 | -36.48 | -48.94 | -26.48 |
|          | (0.07)  | (0.00) | (0.70) | (0.00) | (0.00) | (0.00) | (0.07) | (0.19) | (0.04) | (0.04) | (0.04) | (0.07) | (0.22) |
| RO       | -24.37  | -50.16 | -27.73 | -3.94  | -24.97 | -21.57 | -20.70 | -38.58 | -18.56 | -20.19 | -23.11 | -23.85 | -19.05 |
|          | (0.00)  | (0.00) | (0.00) | (0.70) | (0.07) | (0.07) | (0.11) | (0.04) | (0.11) | (0.15) | (0.15) | (0.00) | (0.07) |
| SE       | -14.51  | -39.98 | -9.74  | -11.10 | -21.30 | -8.46  | -27.00 | -23.06 | -19.46 | -26.28 | -3.76  | 4.43   | 11.59  |
|          | (0.33)  | (0.11) | (0.33) | (0.33) | (0.15) | (0.37) | (0.15) | (0.26) | (0.15) | (0.07) | (0.74) | (0.70) | (0.44) |
| SI       | -22.40  | -58.62 | -31.38 | -20.98 | -34.31 | -30.05 | 76.12  | -23.56 | -17.01 | -24.16 | -23.08 | -38.32 | -43.43 |
|          | (0.52)  | (0.19) | (0.37) | (0.48) | (0.19) | (0.41) | (0.11) | (0.48) | (0.59) | (0.56) | (0.41) | (0.19) | (0.26) |
| SK       | 2.90    | -43.61 | -20.45 | 5.33   | -5.85  | 1.79   | 60.92  | 27.78  | 9.64   | -0.18  | -8.53  | 0.74   | 7.23   |
|          | (0.52)  | (0.19) | (0.33) | (0.81) | (0.63) | (0.85) | (0.00) | (0.26) | (0.78) | (0.96) | (0.70) | (0.96) | (0.85) |
| Total    | -11.85  | -42.76 | -10.88 | -8.46  | -12.85 | -5.21  | -6.67  | -7.37  | -11.08 | -9.33  | -10.96 | -10.14 | -6.47  |
|          | (0.04)  | (0.00) | (0.04) | (0.07) | (0.07) | (0.61) | (0.29) | (0.43) | (0.29) | (0.32) | (0.29) | (0.25) | (0.50) |

The table reports the SC treatment effects and p-values (in parenthesis) of UK exports for each EU country and month of 2021.

Table 20: Disaggregated SC results for UK imports

| Exporter | Jan-Dec          | Jan              | Feb              | Mar              | Apr              | May              | Jun              | Jul              | Aug              | Sep              | Oct              | Nov              | Dec              |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| AT       | -22.91<br>(0.04) | -28.03<br>(0.00) | -9.83<br>(0.30)  | -11.13<br>(0.22) | -23.59<br>(0.00) | -33.50<br>(0.00) | -21.52<br>(0.04) | -22.83<br>(0.04) | -17.46<br>(0.07) | -26.08<br>(0.07) | -33.69<br>(0.00) | -21.48<br>(0.04) | -25.77<br>(0.07) |
| BE       | -23.30<br>(0.19) | -31.80<br>(0.04) | -23.62<br>(0.04) | -22.19<br>(0.11) | -20.11<br>(0.07) | -25.20<br>(0.15) | -18.50<br>(0.22) | -17.21<br>(0.26) | -14.60<br>(0.48) | -20.37<br>(0.30) | -25.18<br>(0.11) | -23.90<br>(0.22) | -36.88<br>(0.07) |
| BG       | -20.97<br>(0.41) | -9.31<br>(0.70)  | -0.64<br>(0.96)  | 14.59<br>(0.59)  | -21.35<br>(0.37) | -21.58<br>(0.44) | -26.72<br>(0.26) | -43.60<br>(0.15) | -37.74<br>(0.15) | -31.54<br>(0.26) | -33.30<br>(0.26) | -15.85<br>(0.78) | -24.57<br>(0.56) |
| CY       | -35.48<br>(0.22) | -32.33<br>(0.15) | -34.70<br>(0.15) | -35.53<br>(0.07) | -37.58<br>(0.07) | -9.33<br>(0.67)  | -28.26<br>(0.22) | -37.87<br>(0.07) | -24.65<br>(0.15) | -41.33<br>(0.04) | -29.71<br>(0.22) | -55.57<br>(0.22) | -58.90<br>(0.00) |
| CZ       | -30.90<br>(0.00) | -30.39<br>(0.00) | -27.73<br>(0.00) | -28.71<br>(0.00) | -41.16<br>(0.00) | -30.82<br>(0.00) | -29.98<br>(0.00) | -35.00<br>(0.00) | -25.24<br>(0.04) | -5.21<br>(0.63)  | -36.20<br>(0.00) | -32.44<br>(0.00) | -47.93<br>(0.00) |
| DE       | -19.88<br>(0.04) | -33.30<br>(0.00) | -18.37<br>(0.04) | -11.40<br>(0.11) | -20.36<br>(0.07) | -18.09<br>(0.04) | -16.55<br>(0.07) | -17.71<br>(0.07) | -14.90<br>(0.26) | -18.13<br>(0.07) | -20.15<br>(0.11) | -22.72<br>(0.00) | -26.92<br>(0.04) |
| DK       | -28.09<br>(0.67) | -31.72<br>(0.41) | -12.87<br>(0.70) | -40.99<br>(0.15) | -26.27<br>(0.37) | -31.98<br>(0.33) | -3.27<br>(0.81)  | 3.25<br>(0.96)   | -46.56<br>(0.19) | -25.44<br>(0.37) | -33.54<br>(0.30) | -38.31<br>(0.56) | -49.38<br>(0.33) |
| EE       | 19.44<br>(0.44)  | -6.81<br>(0.78)  | -2.98<br>(0.85)  | 1.56<br>(0.89)   | 97.92<br>(0.04)  | 28.60<br>(0.48)  | 38.06<br>(0.26)  | -3.04<br>(0.93)  | 10.89<br>(0.81)  | 17.74<br>(0.59)  | 67.98<br>(0.11)  | -15.17<br>(0.70) | -1.49<br>(0.96)  |
| ES       | -19.88<br>(0.11) | -7.53<br>(0.59)  | -11.78<br>(0.22) | -14.78<br>(0.15) | -21.46<br>(0.11) | -23.19<br>(0.07) | -26.45<br>(0.07) | -26.28<br>(0.11) | -14.20<br>(0.41) | -18.56<br>(0.26) | -26.54<br>(0.04) | -19.22<br>(0.26) | -28.55<br>(0.15) |
| FI       | -32.40<br>(0.11) | 2.19<br>(0.81)   | -22.61<br>(0.15) | -39.25<br>(0.07) | -45.17<br>(0.00) | -48.25<br>(0.00) | -34.80<br>(0.15) | -33.70<br>(0.07) | -37.97<br>(0.00) | -35.51<br>(0.11) | -28.91<br>(0.07) | -21.82<br>(0.19) | -43.06<br>(0.19) |
| FR       | -22.25<br>(0.11) | -31.81<br>(0.04) | -23.65<br>(0.07) | -21.38<br>(0.07) | -27.23<br>(0.07) | -25.84<br>(0.15) | -23.25<br>(0.19) | -19.28<br>(0.22) | -12.87<br>(0.33) | -14.93<br>(0.15) | -17.50<br>(0.30) | -23.50<br>(0.15) | -25.76<br>(0.15) |
| GR       | -24.58<br>(0.00) | -12.43<br>(0.19) | -31.16<br>(0.00) | -21.38<br>(0.04) | -36.78<br>(0.00) | -34.41<br>(0.04) | -32.99<br>(0.00) | -17.42<br>(0.15) | -8.85<br>(0.26)  | -28.34<br>(0.11) | -30.31<br>(0.04) | -20.33<br>(0.04) | -20.53<br>(0.11) |

The table reports the SC treatment effects and p-values (in parenthesis) of UK imports for each EU country and month of 2021.

Table 21: Disaggregated SC results for UK imports (continued)

| Exporter | Jan-Dec          | Jan              | Feb              | Mar              | Apr              | May              | Jun              | Jul              | Aug              | Sep              | Oct              | Nov              | Dec              |
|----------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| HR       | -53.57<br>(0.07) | -73.38<br>(0.00) | -51.27<br>(0.04) | -63.97<br>(0.04) | -37.01<br>(0.11) | -70.30<br>(0.04) | -49.38<br>(0.04) | -29.78<br>(0.11) | -56.94<br>(0.04) | -52.22<br>(0.15) | -50.75<br>(0.07) | -62.90<br>(0.04) | -44.90<br>(0.11) |
| HU       | -32.86<br>(0.07) | -40.97<br>(0.07) | -26.06<br>(0.07) | -33.71<br>(0.00) | -49.07<br>(0.00) | -40.57<br>(0.00) | -33.01<br>(0.04) | -33.54<br>(0.07) | -26.12<br>(0.11) | -24.68<br>(0.07) | -30.62<br>(0.07) | -23.95<br>(0.19) | -32.02<br>(0.11) |
| IE       | -19.18<br>(0.19) | -10.50<br>(0.33) | -36.04<br>(0.00) | -17.97<br>(0.22) | -23.02<br>(0.11) | -20.73<br>(0.04) | -5.13<br>(0.59)  | -5.55<br>(0.56)  | -27.65<br>(0.04) | -19.74<br>(0.26) | -33.80<br>(0.04) | -5.72<br>(0.67)  | -24.35<br>(0.07) |
| IT       | -16.89<br>(0.04) | -32.05<br>(0.00) | -19.00<br>(0.07) | -10.49<br>(0.11) | -14.65<br>(0.04) | -14.37<br>(0.07) | -15.51<br>(0.04) | -17.66<br>(0.00) | -3.57<br>(0.56)  | -18.38<br>(0.00) | -16.18<br>(0.04) | -18.45<br>(0.04) | -22.31<br>(0.00) |
| LT       | -10.24<br>(0.81) | 10.49<br>(0.52)  | -15.53<br>(0.44) | -22.16<br>(0.19) | -1.65<br>(0.85)  | -18.38<br>(0.41) | -11.19<br>(0.41) | -2.68<br>(0.81)  | -0.89<br>(0.93)  | -11.61<br>(0.67) | -15.56<br>(0.52) | -15.61<br>(0.41) | -18.15<br>(0.22) |
| LU       | -40.16<br>(0.37) | -47.28<br>(0.11) | -41.87<br>(0.00) | -42.44<br>(0.19) | -55.58<br>(0.04) | -38.86<br>(0.22) | -37.45<br>(0.41) | -34.71<br>(0.37) | -36.40<br>(0.11) | -28.32<br>(0.41) | -30.87<br>(0.41) | -32.83<br>(0.33) | -55.32<br>(0.22) |
| LV       | 28.95<br>(0.15)  | 1.43<br>(0.93)   | 2.48<br>(0.85)   | 46.83<br>(0.04)  | 5.32<br>(0.81)   | 8.91<br>(0.74)   | 33.41<br>(0.11)  | 57.55<br>(0.11)  | 79.16<br>(0.04)  | 65.52<br>(0.07)  | 30.54<br>(0.15)  | 17.19<br>(0.59)  | -0.89<br>(0.96)  |
| MT       | -34.26<br>(0.85) | -21.27<br>(0.70) | -21.49<br>(0.74) | -16.06<br>(0.74) | -45.99<br>(0.30) | -27.27<br>(0.52) | 2.22<br>(0.93)   | -22.30<br>(0.78) | -38.30<br>(0.67) | -47.87<br>(0.48) | -70.91<br>(0.26) | -28.77<br>(0.59) | -73.13<br>(0.37) |
| NL       | -35.52<br>(0.00) | -32.04<br>(0.04) | -33.71<br>(0.00) | -34.94<br>(0.00) | -41.12<br>(0.00) | -39.49<br>(0.00) | -35.68<br>(0.04) | -32.18<br>(0.04) | -34.40<br>(0.00) | -32.47<br>(0.04) | -35.48<br>(0.04) | -34.18<br>(0.04) | -40.57<br>(0.04) |
| PL       | -25.30<br>(0.00) | -26.86<br>(0.00) | -26.06<br>(0.04) | -21.95<br>(0.00) | -25.53<br>(0.00) | -23.63<br>(0.04) | -24.52<br>(0.07) | -25.19<br>(0.04) | -23.56<br>(0.04) | -21.26<br>(0.00) | -25.76<br>(0.00) | -27.83<br>(0.04) | -31.42<br>(0.04) |
| PT       | -25.00<br>(0.00) | -16.25<br>(0.15) | -21.56<br>(0.00) | -18.16<br>(0.11) | -36.49<br>(0.04) | -32.27<br>(0.04) | -24.66<br>(0.07) | -40.46<br>(0.00) | -16.20<br>(0.22) | 1.22<br>(0.85)   | -34.30<br>(0.00) | -18.22<br>(0.15) | -42.71<br>(0.00) |
| RO       | -22.20<br>(0.11) | -34.56<br>(0.00) | -26.82<br>(0.04) | -9.23<br>(0.63)  | -19.60<br>(0.15) | -13.70<br>(0.52) | -7.74<br>(0.59)  | -9.77<br>(0.63)  | 8.07<br>(0.74)   | -26.23<br>(0.11) | -37.62<br>(0.00) | -41.14<br>(0.11) | -48.08<br>(0.04) |
| SE       | -25.05<br>(0.07) | -17.82<br>(0.22) | -20.64<br>(0.04) | -18.95<br>(0.07) | -34.52<br>(0.11) | -28.76<br>(0.00) | -15.12<br>(0.19) | -18.96<br>(0.15) | -17.00<br>(0.19) | -15.56<br>(0.22) | -34.11<br>(0.00) | -38.31<br>(0.00) | -40.82<br>(0.04) |
| SI       | -11.31<br>(0.59) | -10.33<br>(0.67) | -4.52<br>(0.78)  | 1.23<br>(0.81)   | -27.57<br>(0.19) | -10.68<br>(0.48) | -1.98<br>(0.89)  | -18.40<br>(0.37) | -10.43<br>(0.30) | -17.51<br>(0.48) | -13.84<br>(0.37) | -6.94<br>(0.70)  | -14.78<br>(0.48) |
| SK       | -26.77<br>(0.19) | -53.70<br>(0.04) | -32.79<br>(0.07) | -3.79<br>(0.67)  | -36.71<br>(0.07) | -22.29<br>(0.19) | -5.09<br>(0.70)  | -26.34<br>(0.37) | -39.60<br>(0.07) | -17.28<br>(0.41) | -30.13<br>(0.15) | -21.31<br>(0.26) | -32.22<br>(0.30) |
| Total    | -24.05<br>(0.00) | -28.48<br>(0.00) | -23.24<br>(0.00) | -19.89<br>(0.00) | -26.42<br>(0.00) | -25.40<br>(0.00) | -20.99<br>(0.00) | -20.80<br>(0.00) | -19.81<br>(0.04) | -20.64<br>(0.04) | -26.06<br>(0.00) | -24.54<br>(0.00) | -32.33<br>(0.00) |

The table reports the SC treatment effects and p-values (in parenthesis) of UK imports for each EU country and month of 2021.

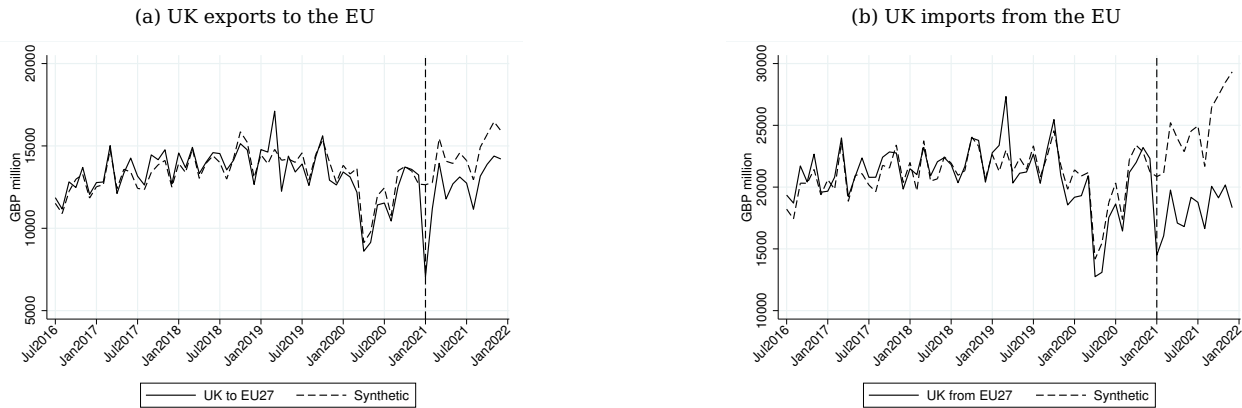
## A.7 Alternative predictors for SC

As a robustness to our SC estimation, we changed the set of predictors. Rather than matching over the entire pre-treatment period (Jul2016-Dec2020) we use as predictors the average of the outcome variable over each month. This gives us 12 predictors that will be the average over January 2017-20, the average over February 2017-20, March, ... The resulting series are plotted in figure 14 for both exports and imports. This robustness gives a cumulative effect over January-December 2021 of -14.2% (p-value from permutation 0.04) for exports



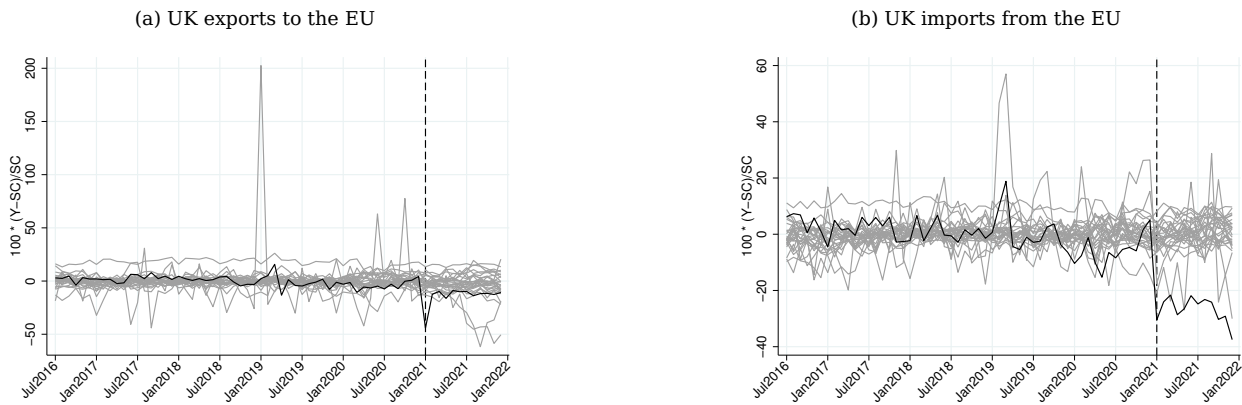
and -25% for imports (p-value=0.00), slightly larger than our baseline estimates. The permutation tests also show similar significance levels.

Figure 14: UK-EU trade SCM result



The figure reports the SC estimates for UK exports to and imports from the EU27. The pre-treatment period is Jul2016-Dec2020, matching is performed over month averages of the pre-treatment values of the outcome variable.

Figure 15: UK-EU trade SCM placebo result



The figure reports the SC estimates for UK exports to and imports from the EU27. The pre-treatment period is Jul2016-Dec2020, matching is performed over month averages of the pre-treatment values of the outcome variable.

## A.8 Synthetic Control: resampling inference

In this section we replicate the inference method developed by Saia (2017) and adopted by Douch and Edwards (2021) in their analysis of post referendum UK-EU trade. The method computes standard errors of SC treatment effect based on resampling. Given a donor pool of  $N$  units, take a random sample from donor units

of size  $z \times N$  (with  $z < 1$ ) and compute the SC. Repeat the operation  $M$  times to obtain  $M$  synthetic controls. The final synthetic control is given by the average SC across the  $M$  iterations, and the treatment effect is computed as:

$$\hat{\alpha}_j = \frac{1}{M} \sum_{m=1}^M \frac{\sum_{t \in T_1} (Y_{jt} - \hat{Y}_{jt,m})}{\sum_{t \in T_1} \hat{Y}_{jt,m}} \quad (15)$$

where  $j$  indicates the trade partner of the UK (e.g., France) and  $T_1$  is the treatment period. Then  $Y_{jt}$  is the observed outcome and  $\hat{Y}_{jt,m}$  is the  $m$ th SC for country  $j$ . Saia (2017) computes the standard deviation of the treatment effect as:

$$\sigma_j = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{\alpha}_j - \hat{\alpha}_{j,m})^2} \quad (16)$$

We can compute similar statistics for total UK-EU trade by aggregating the individual  $j$  countries SCs:

$$\hat{\alpha}_{EU} = \frac{1}{M} \sum_{m=1}^M \frac{\sum_{t \in T_1} (\sum_j Y_{jt} - \sum_j \hat{Y}_{jt,m})}{\sum_{t \in T_1} \sum_j \hat{Y}_{jt,m}} \quad (17)$$

Together with this we can also compute the standard deviation for EU total treatment effect:

$$\sigma_{EU} = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{\alpha}_{EU} - \hat{\alpha}_{EU,m})^2} \quad (18)$$

We compute these statistics for total UK-EU exports and imports. We follow Saia (2017) in setting  $M = 500$ . For the resampling size, Saia (2017) selects 5/6 observations for each iteration. We consider three different resample sizes. Given a donor pool of 60 units, we draw either 50, 40 or 32 units. For each resample size we perform 500 iterations and compute average treatment effect and standard errors as in equations (17) and (18) respectively.

Results are reported in Table 22. As we can see the method yields very small standard errors, although with some variation across different resample sizes.

Table 22: SC resampling inference

| Flow                     | Exports |        |        | Imports |        |        |
|--------------------------|---------|--------|--------|---------|--------|--------|
|                          | 32      | 40     | 50     | 32      | 40     | 50     |
| Average treatment effect | -0.102  | -0.110 | -0.116 | -0.236  | -0.238 | -0.238 |
| Standard error           | 0.032   | 0.023  | 0.014  | 0.019   | 0.013  | 0.009  |

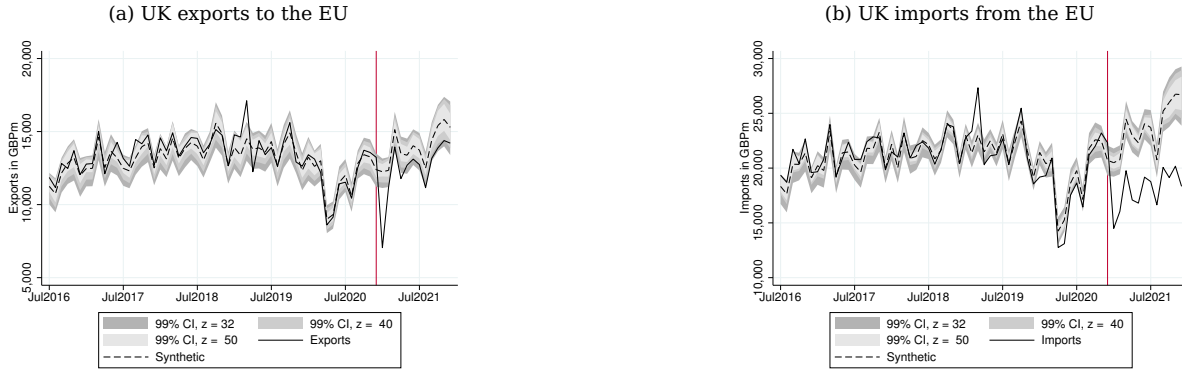
We then compute a confidence interval for the SC in each period based on the resampling method following Saia 2017. First, we compute the standard deviation of the SC:

$$\sigma_{\hat{Y},t} = \sqrt{\frac{1}{M-1} \sum_{m=1}^M (\hat{Y}_{EU,t} - \hat{Y}_{EU,t,m})^2}$$

where  $\hat{Y}_{EU,t} = M^{-1} \sum_{m=1}^M \hat{Y}_{EU,t,m}$  is the average SC. Second, we compute the C.I. taking  $\hat{Y}_{EU,t} \pm \alpha^* \sigma_{\hat{Y},t}$ . Results are reported in Figure 16, where we plot actual and synthetic UK-EU trade together with the 99% confidence interval. As we can see the 99% confidence interval is tight, but different resample sizes give different answers for exports.

For imports (panel b), independently of the resample size (32, 40 or 50) we always find a strong treatment effects. and shows a treatment effect for exports as well as for imports. On the other hand, for exports we definitely find an effect in January 2021 but for other months of 2021 the resampling C.I. with resample size of 32 or 40 often overlap with the actual series.

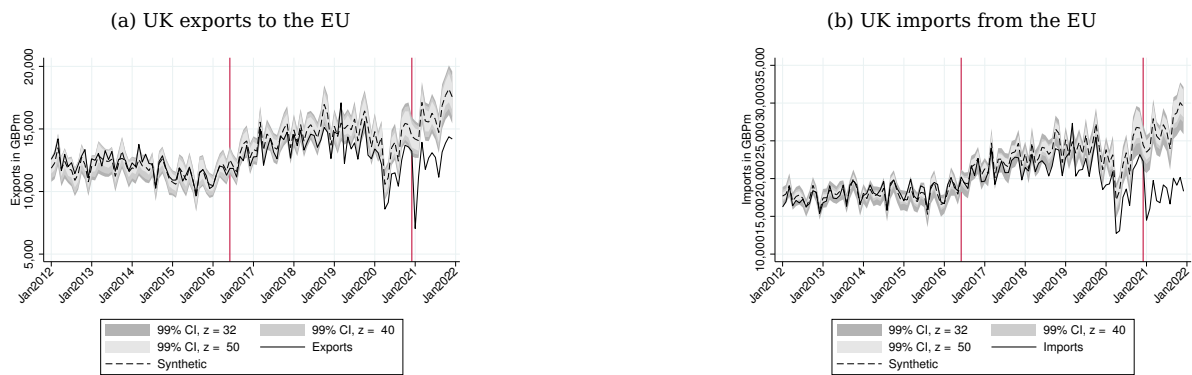
Figure 16: SC resampling confidence interval, Jul2016-Dec2021



The figures plot UK exports to and imports from the EU27 and the average SC. The grey area represents the 99

We then perform the same calculation for the SC matched over the period Jan2012-Jun2016. In this case evidence in support of a referendum effect is scarce, as the actual trade series – either for exports or imports – is often within the C.I. Differently from the previous exercise, the resampling-based inference method would attribute a significant treatment effect to the the TCA for exports, although this is always smaller than the imports effect.

Figure 17: SC resampling confidence interval, Jan2012-Dec2021



The figures plot UK exports to and imports from the EU27 and the average SC. The grey area represents the 99